

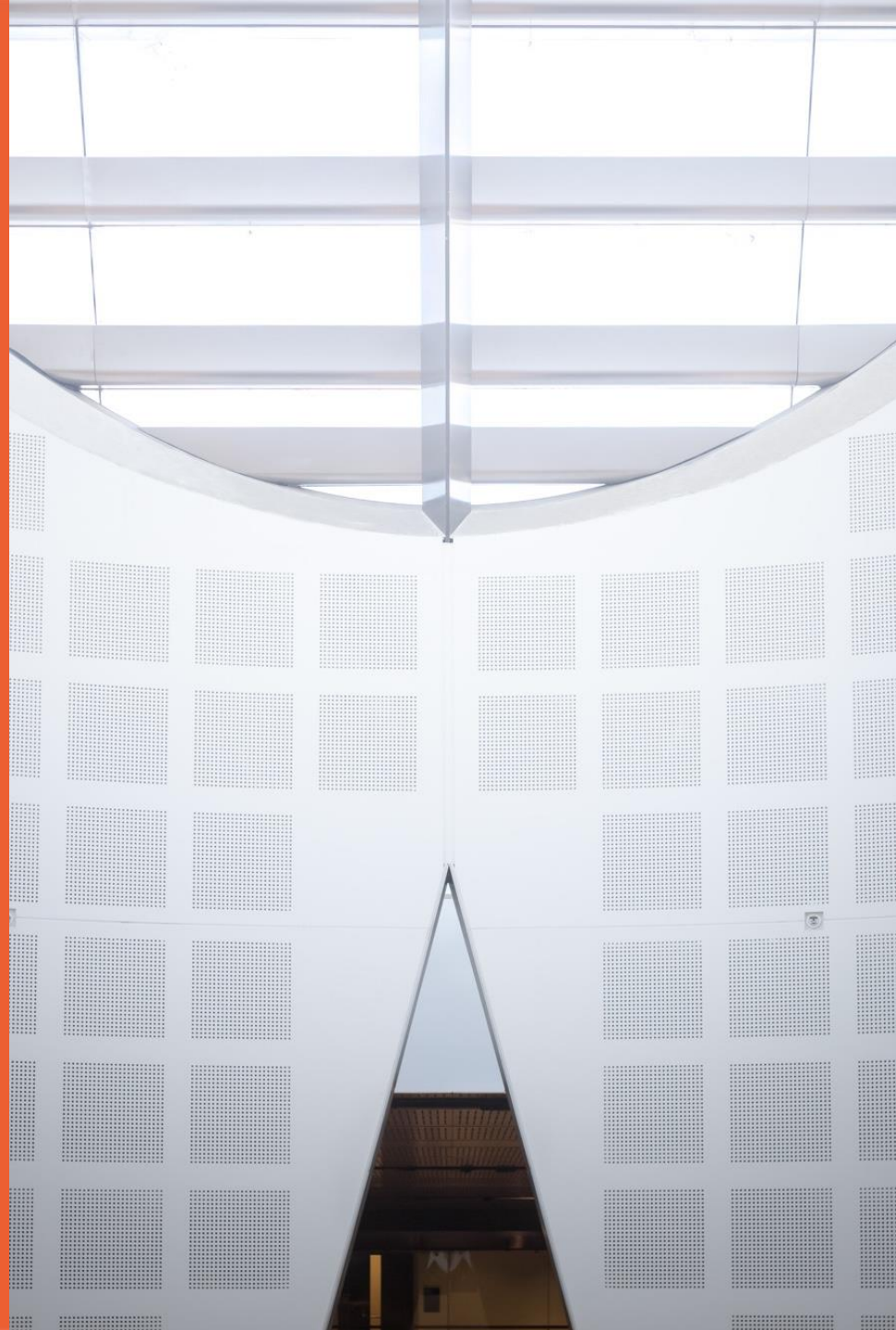
Sequential Modeling Architecture - II

Yunke Wang

School of Computer Science



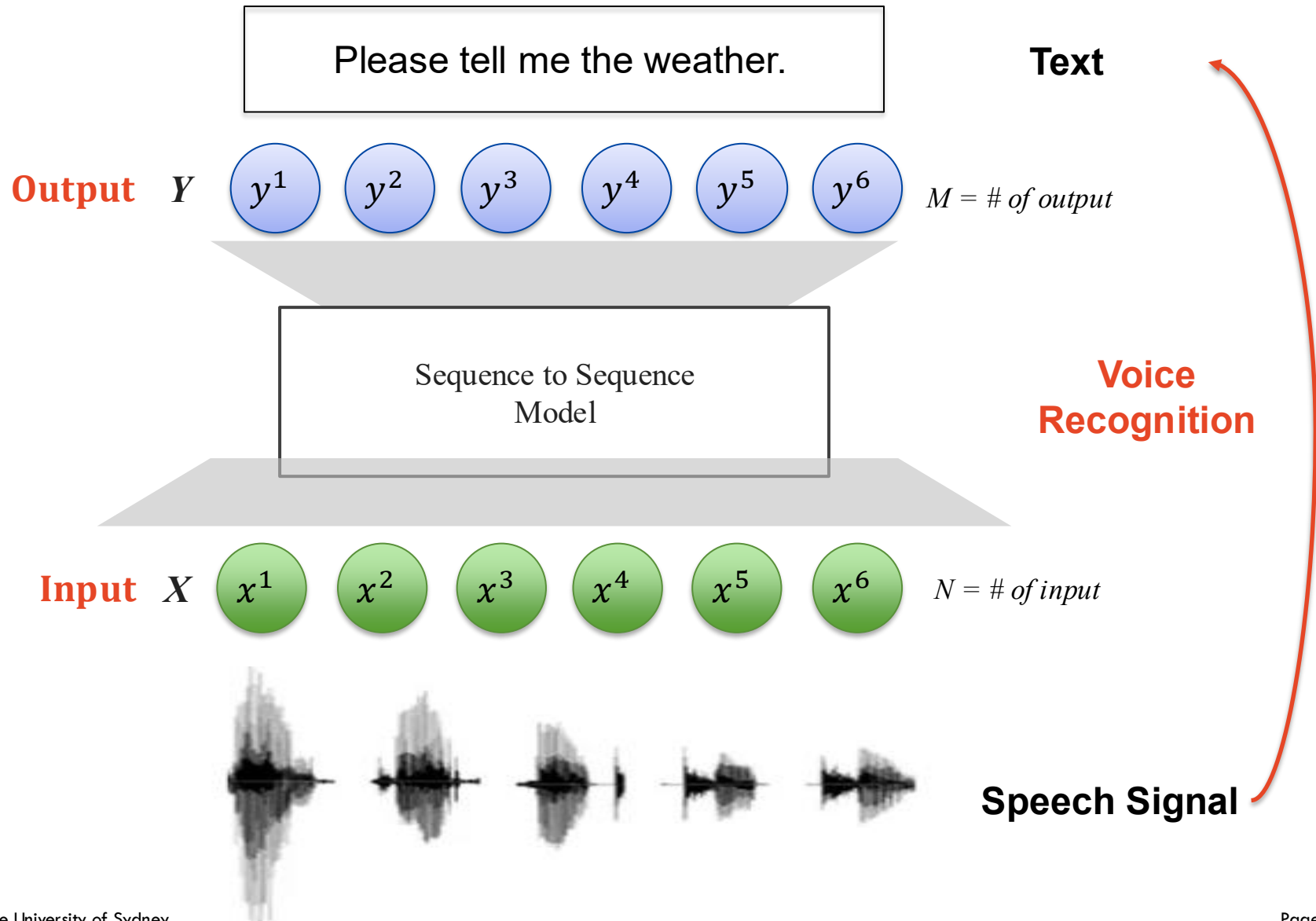
THE UNIVERSITY OF
SYDNEY



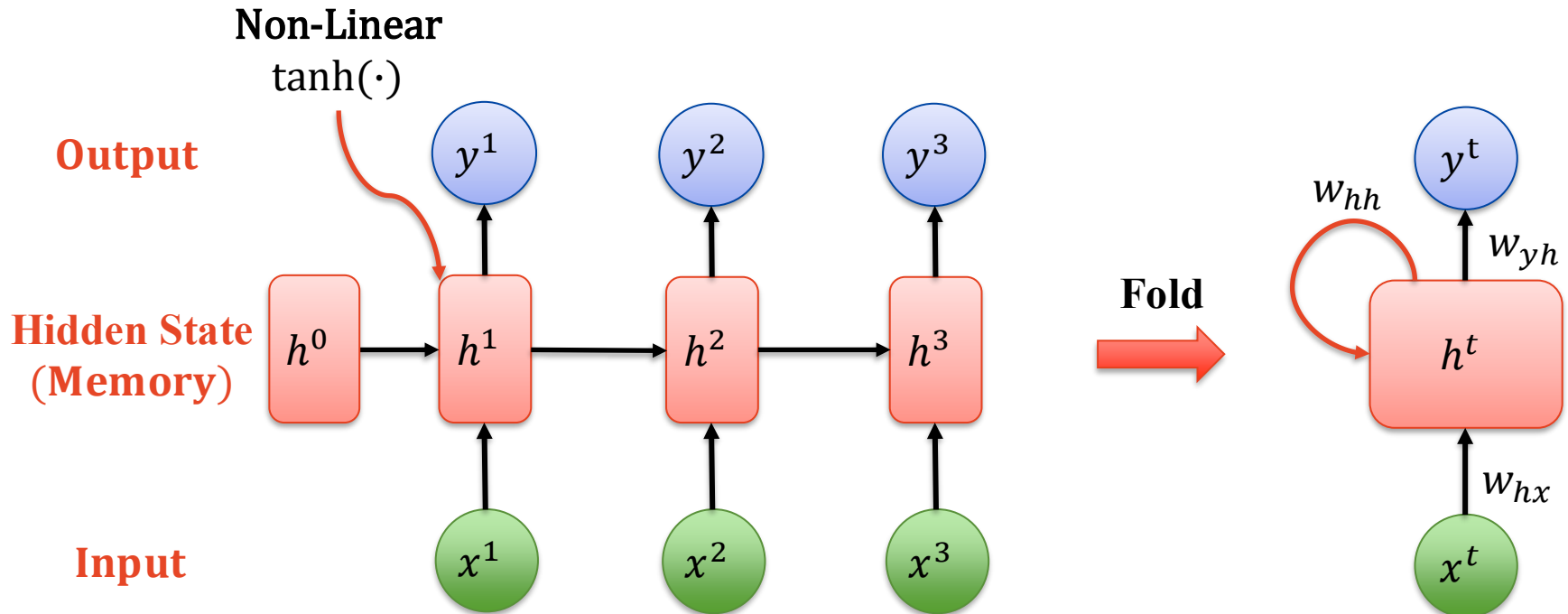
Quick Review

S2S & RNN & Transformer

Sequence to Sequence Model



Recurrent Neural Network



$$\square h_t = \tanh(w_{hh}h_{t-1} + w_{hx}x_t + b_h)$$

$$\square y_t = w_{yh}h_t$$

- Infinite Context Window
- Complexity: $O(N)$
- Not parallelizable

Recurrent Neural Network

□ Vanishing (exploding) gradient problem

□ $h_t = \tanh(w_{hh}h_{t-1} + w_{hx}x_t + b_h)$

□ $y_t = w_{yh}h_t$

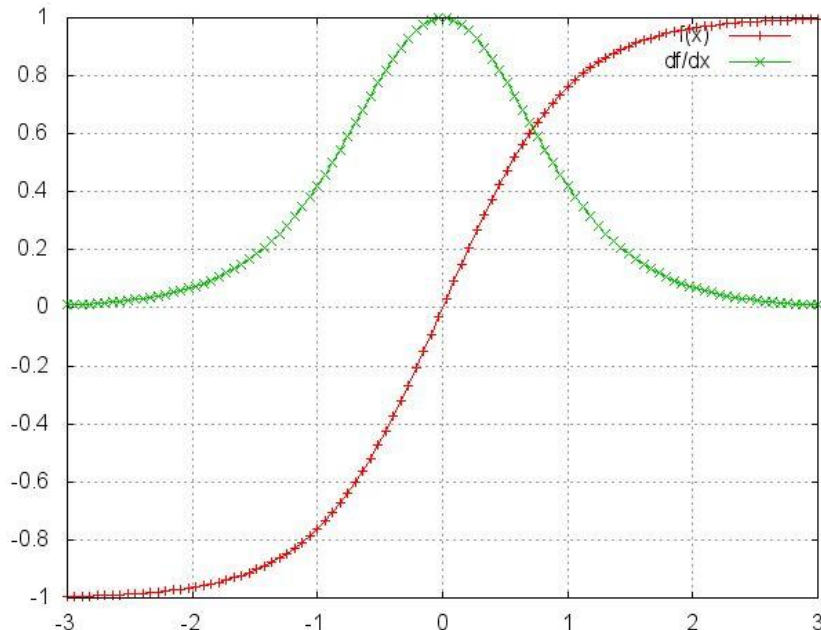
□ For every step, its loss is E_n

$$0.9^{1000} \approx 0$$

$$1.01^{1000} \approx 21000$$

$$\frac{\partial E_3}{\partial w_{hh}} = \frac{\partial E_3}{\partial y_3} \frac{\partial y_3}{\partial h_3} \frac{\partial h_3}{\partial w_{hh}}$$

$$(\tanh)'w_{hh}(\tanh)'w_{hh}(\tanh)'w_{hh} \cdots$$



$$(\tanh)' \leq 1$$

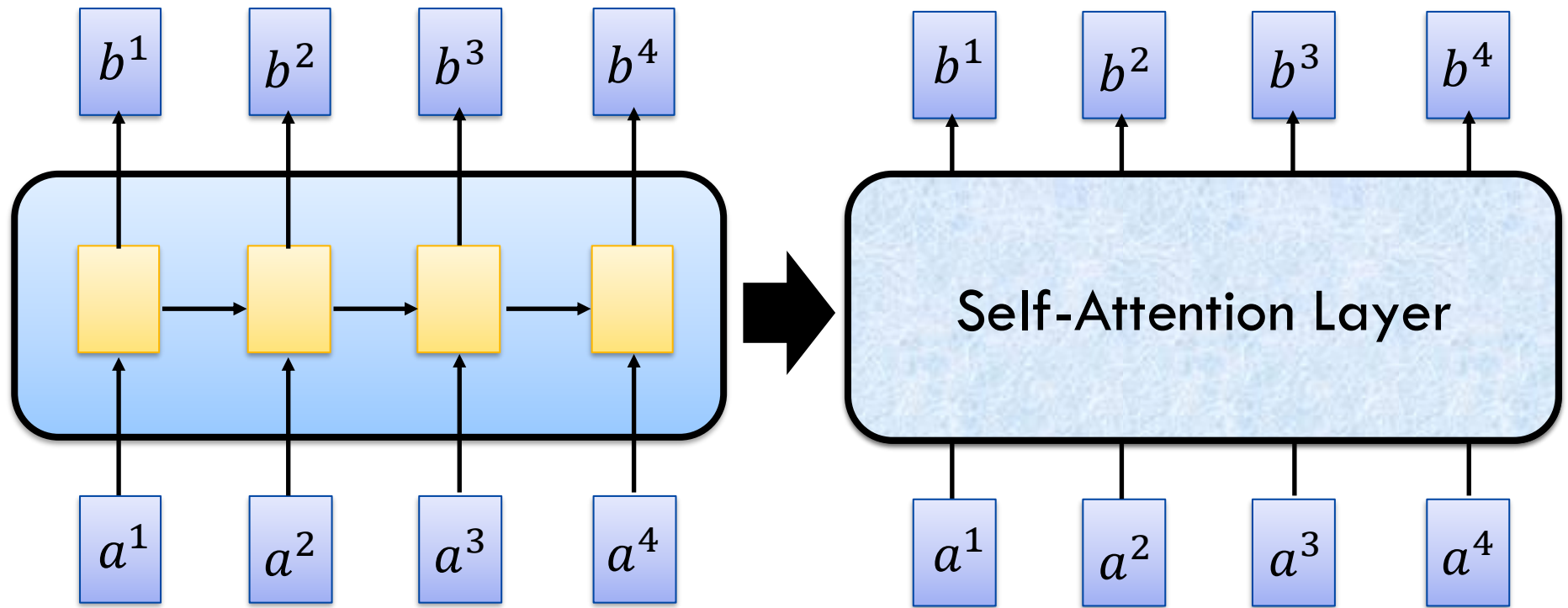
If $0 < w_{hh} < 1$, **Vanishing Gradients**

$$(\tanh)'w_{hh}(\tanh)'w_{hh}(\tanh)'w_{hh} \cdots \rightarrow 0$$

If w_{hh} is larger, **Exploding Gradients**

$$(\tanh)'w_{hh}(\tanh)'w_{hh}(\tanh)'w_{hh} \cdots \rightarrow \infty$$

Self-Attention



You can try to replace any thing that has been done by RNN with self-attention.

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez*[†]
University of Toronto
aidan@cs.toronto.edu

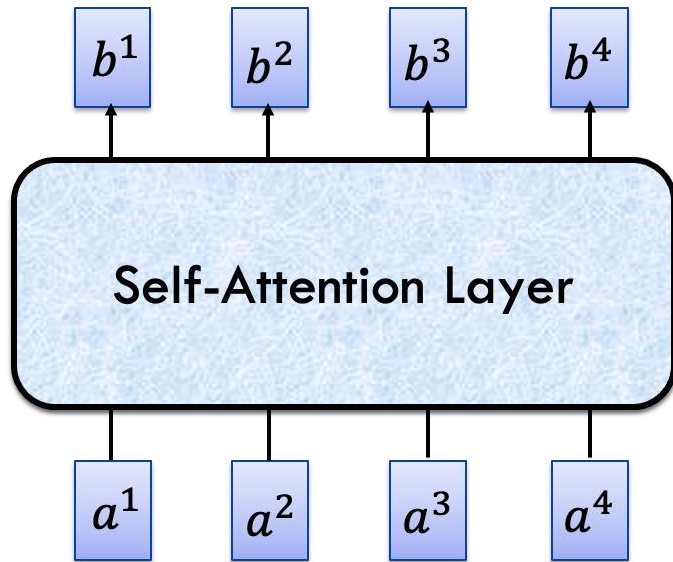
Lukasz Kaiser*
Google Brain
lukasz.kaiser@google.com

Illia Polosukhin*[‡]
illia.polosukhin@gmail.com



Attention is all
you need.

Self-Attention



q : **query** (to match others)

$$q^i = W^q a^i$$

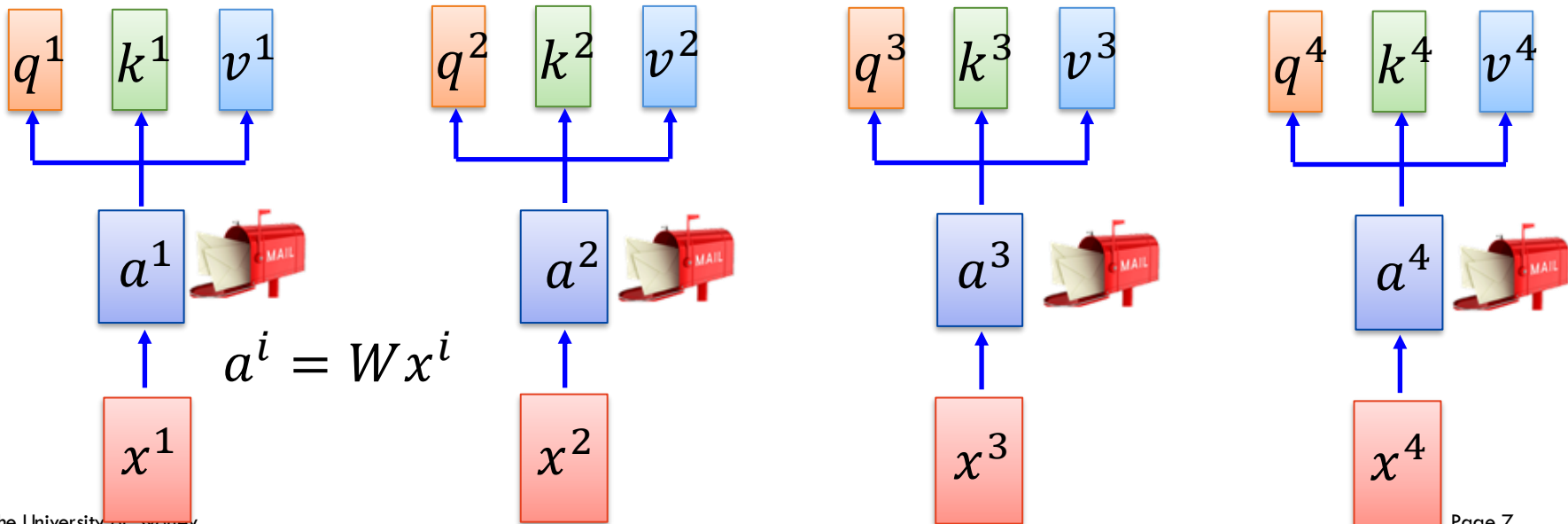


k : **key** (to be matched)

$$k^i = W^k a^i$$

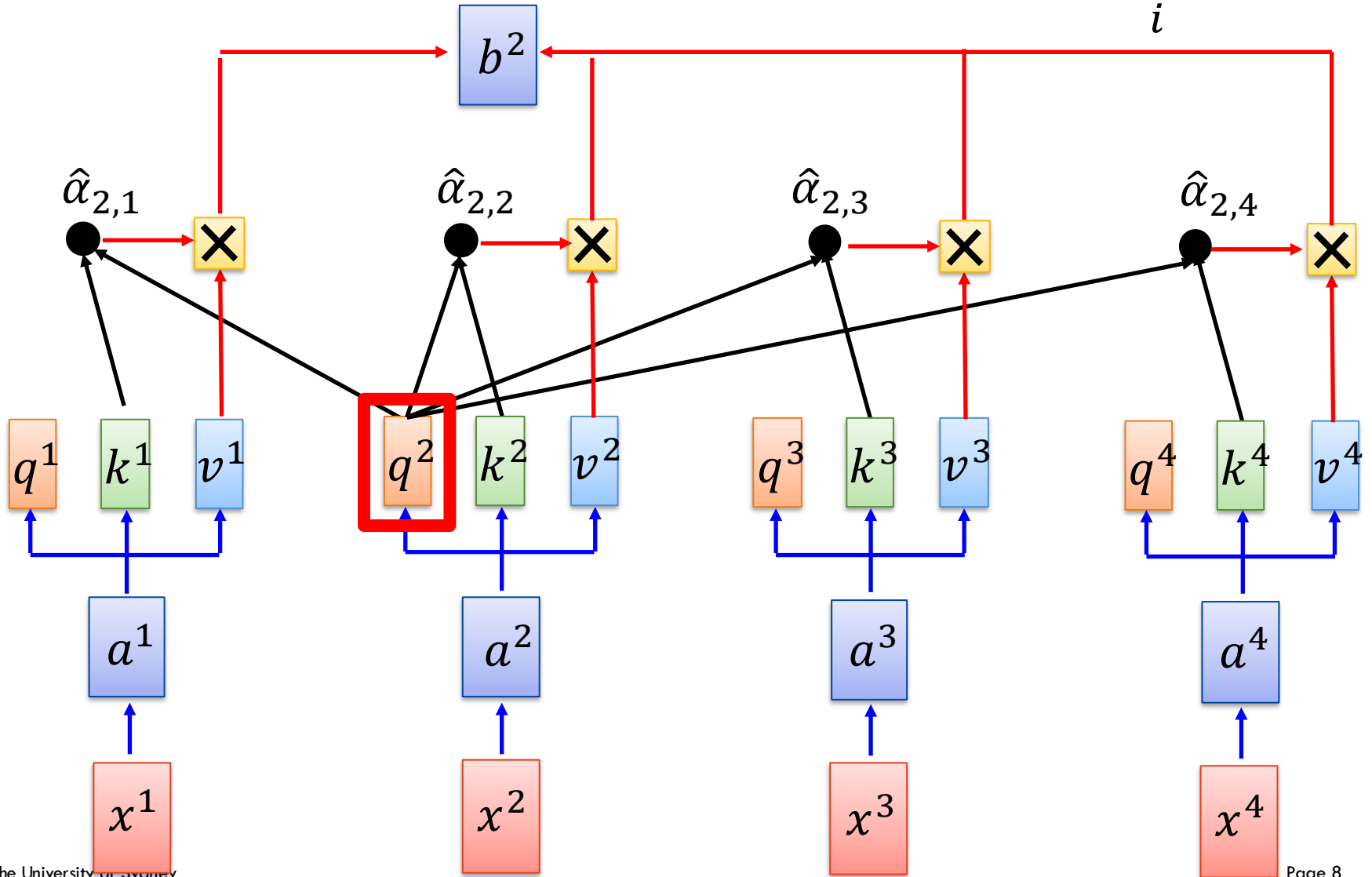
v : **value** (information to be extracted)

$$v^i = W^v a^i$$



Self-Attention

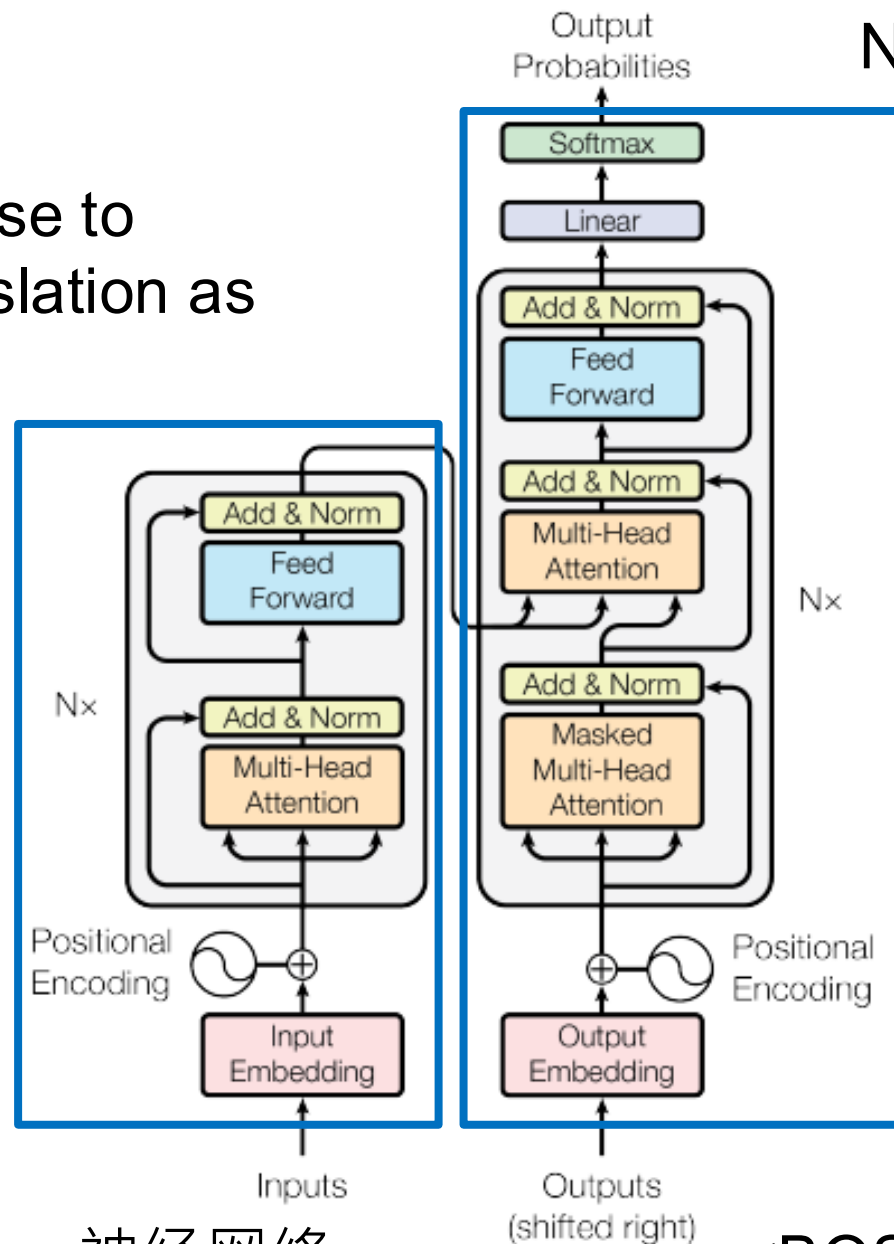
$$b^2 = \sum_i \hat{\alpha}_{2,i} v^i$$



Transformer

Using Chinese to
English translation as
example

Encoder



Neural Network

Decoder

神经网络

Outputs
(shifted right)

<BOS>

BERT

A **transformer** uses Encoder stack to model input, and uses Decoder stack to model output (using input information from encoder side).

If we are only interested in training a language model for the input for some other tasks, then we do not need the Decoder of the transformer, that gives us **BERT**.

1-of-N Encoding

apple = [1 0 0 0 0]

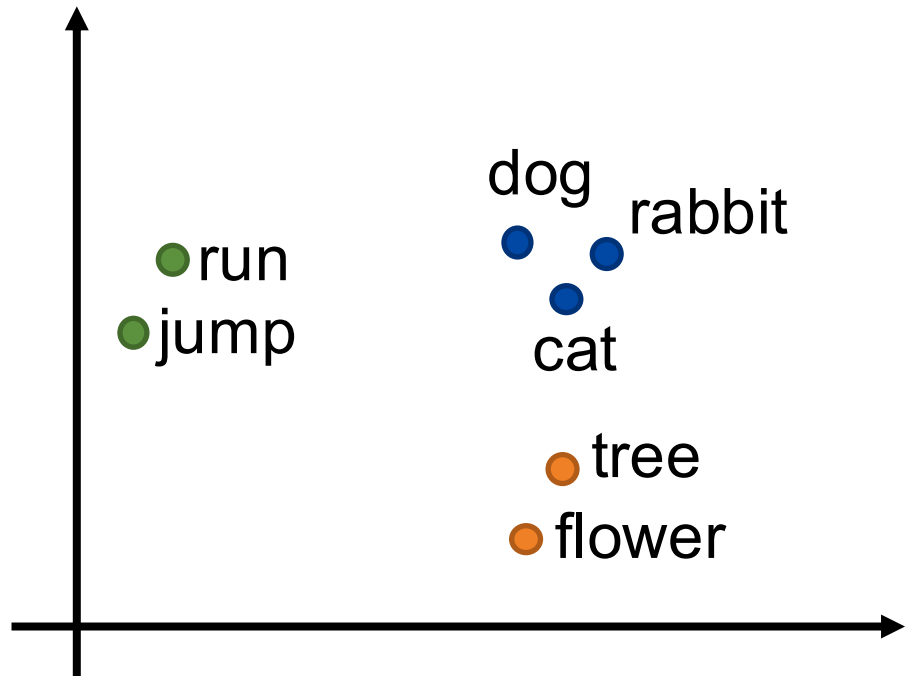
bag = [0 1 0 0 0]

cat = [0 0 1 0 0]

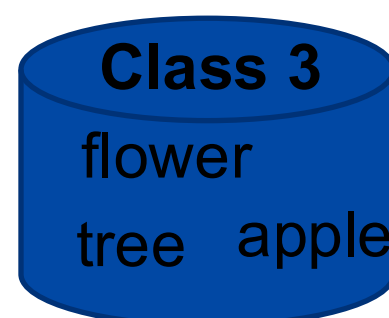
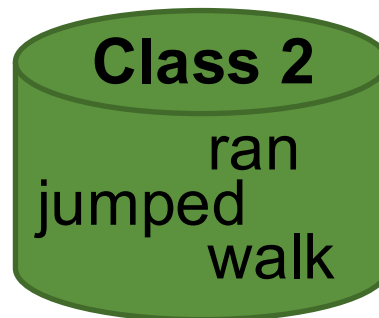
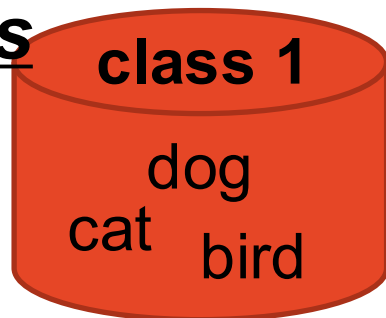
dog = [0 0 0 1 0]

elephant = [0 0 0 0 1]

Word Embedding



Word Class



A word can have multiple senses.

Have you paid that money to the **bank** yet ?

It is safest to deposit your money in the **bank** .

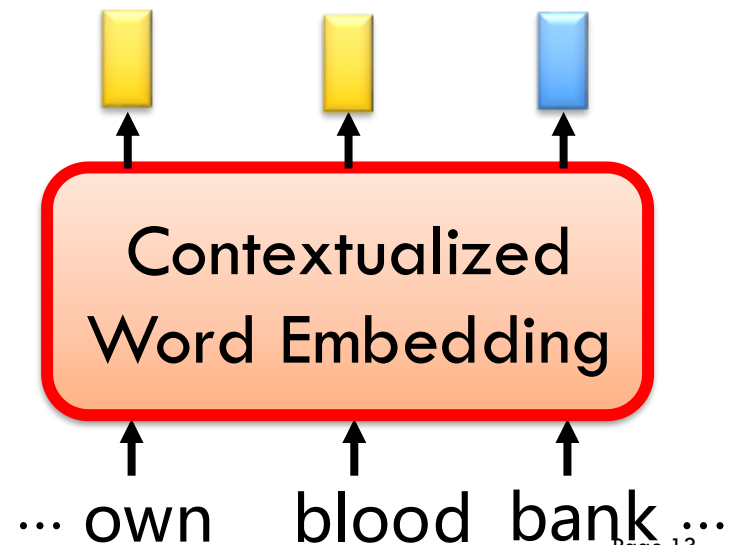
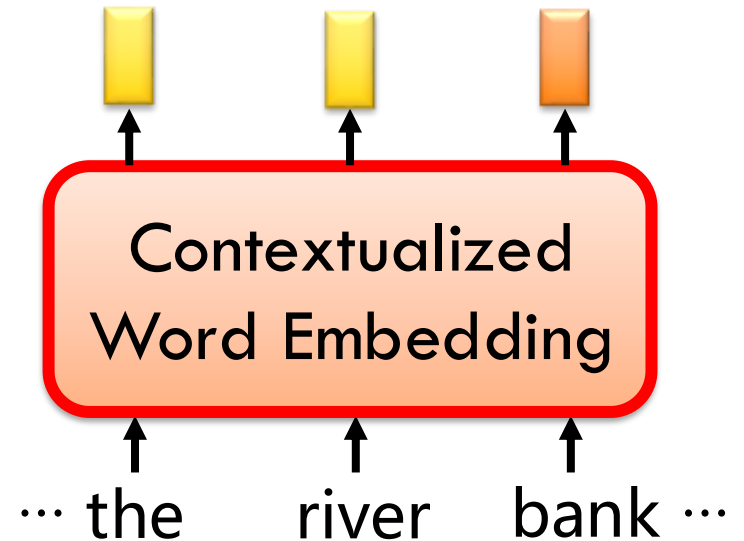
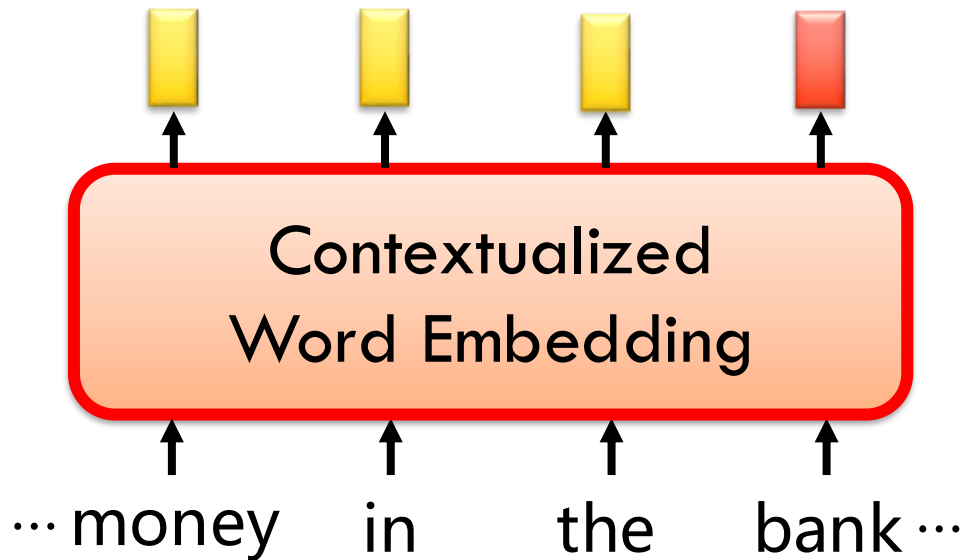
The victim was found lying dead on the river **bank** .

They stood on the river bank to fish.

The hospital has its own blood **bank**.

The third sense or not?

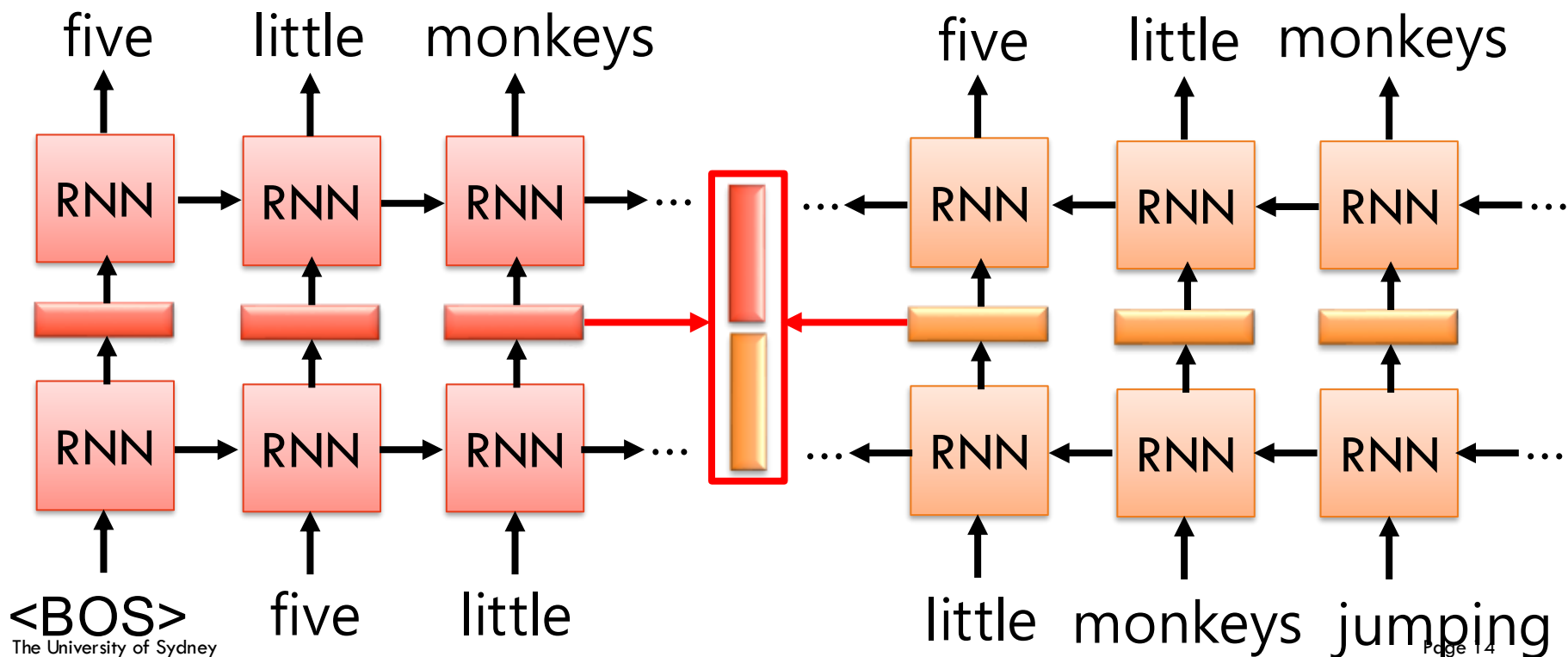
Contextualized Word Embedding



Embeddings from Language Model (ELMO)



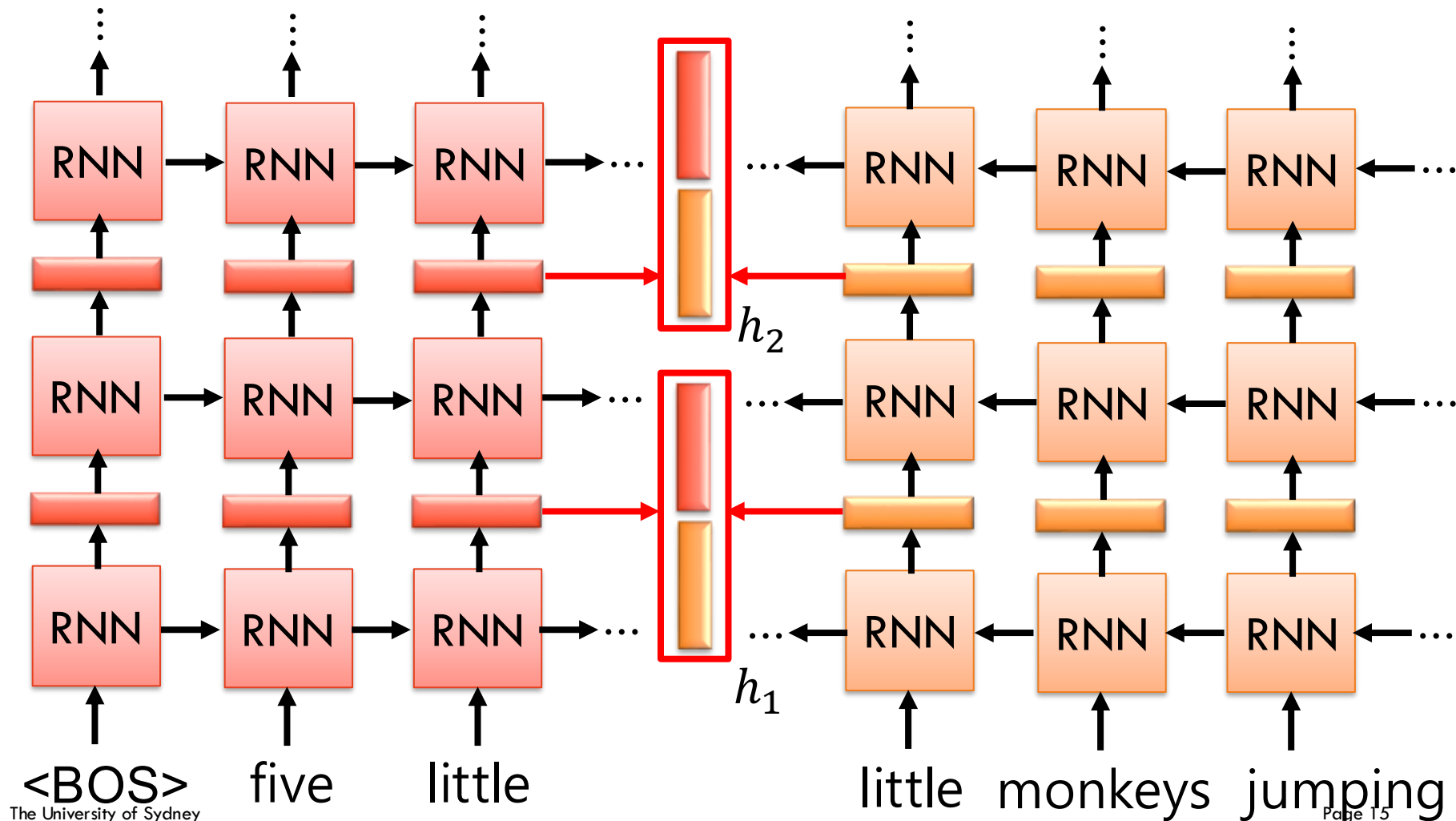
- RNN-based language models (trained from lots of sentences)
e.g. given “five little monkeys jumping on the bed”



ELMO

Each layer in deep LSTM can generate a latent representation.

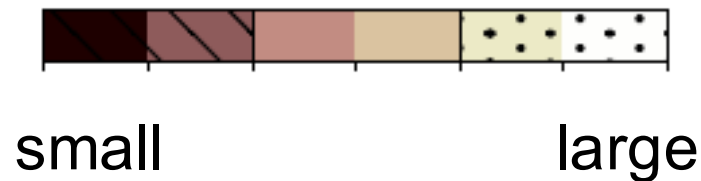
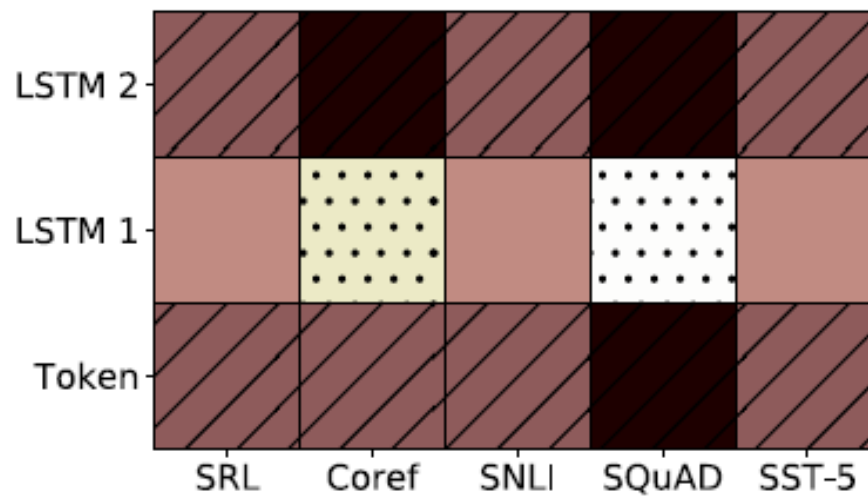
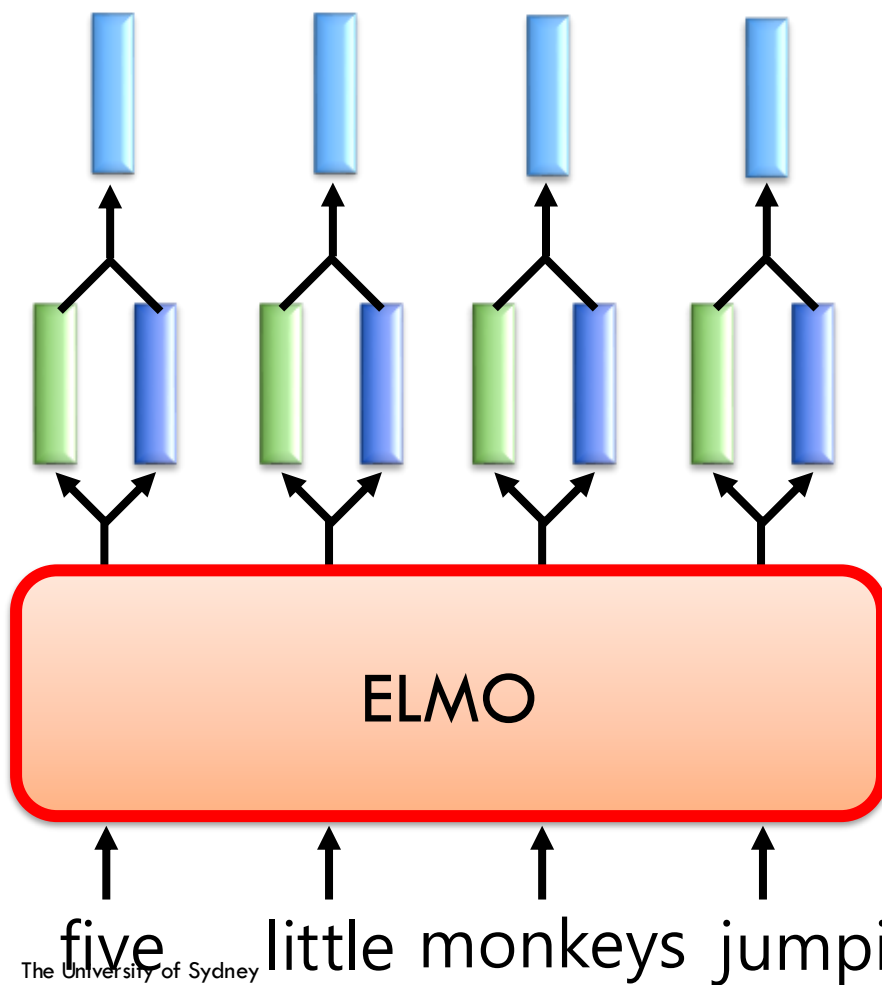
Which one should we use???



ELMO

$$\text{blue bar} = \alpha_1 \text{green bar} + \alpha_2 \text{blue bar}$$

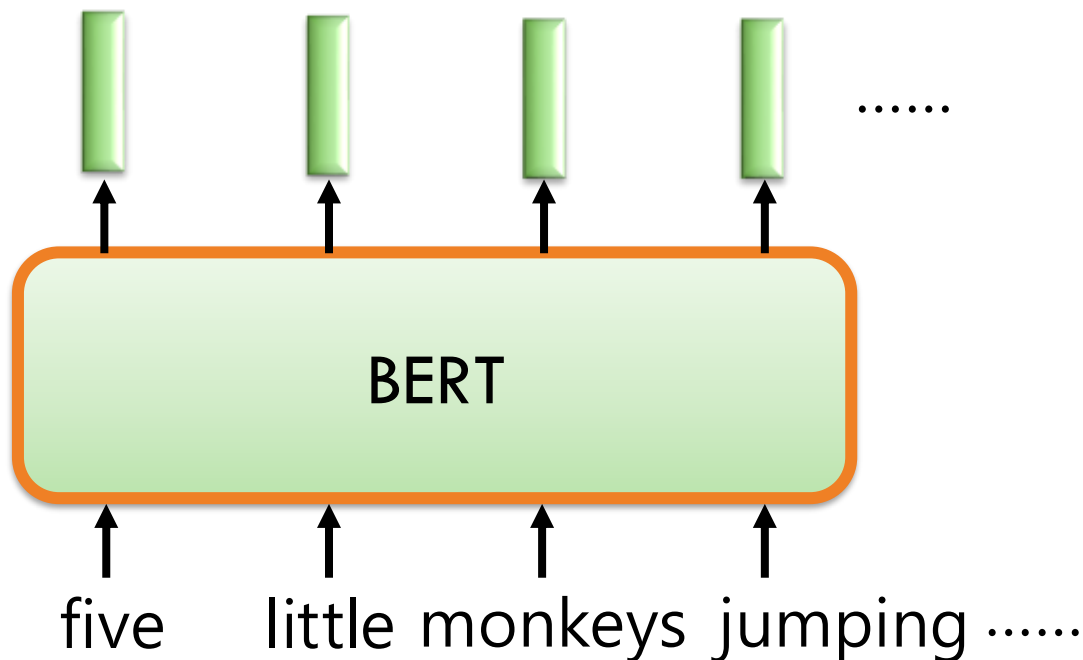
Learned with the down stream tasks



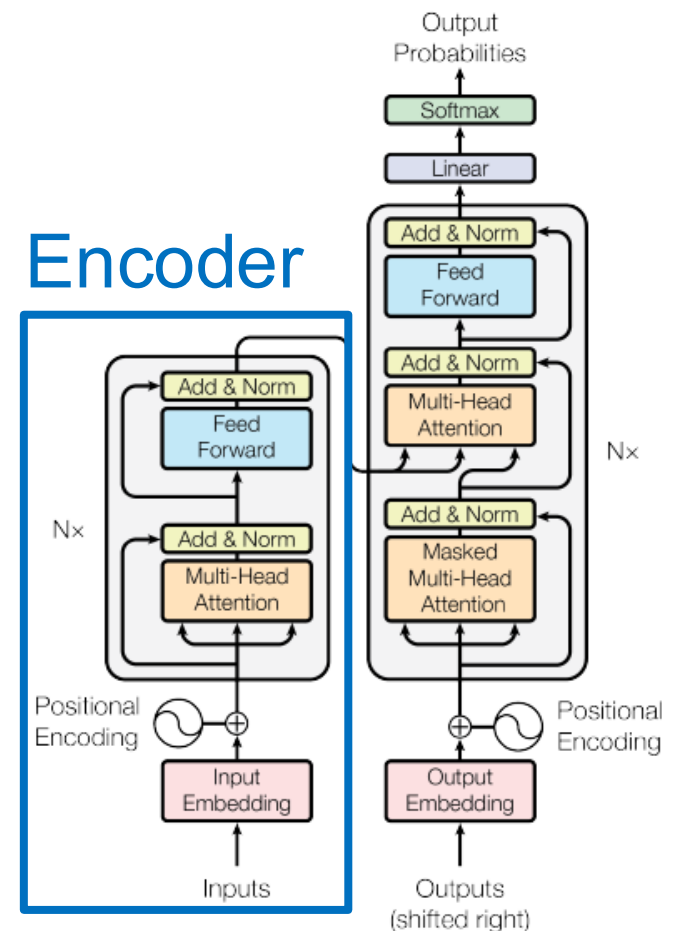
Bidirectional Encoder Representations from Transformers (BERT)

- BERT = Encoder of Transformer

Learned from a large amount of text without annotation

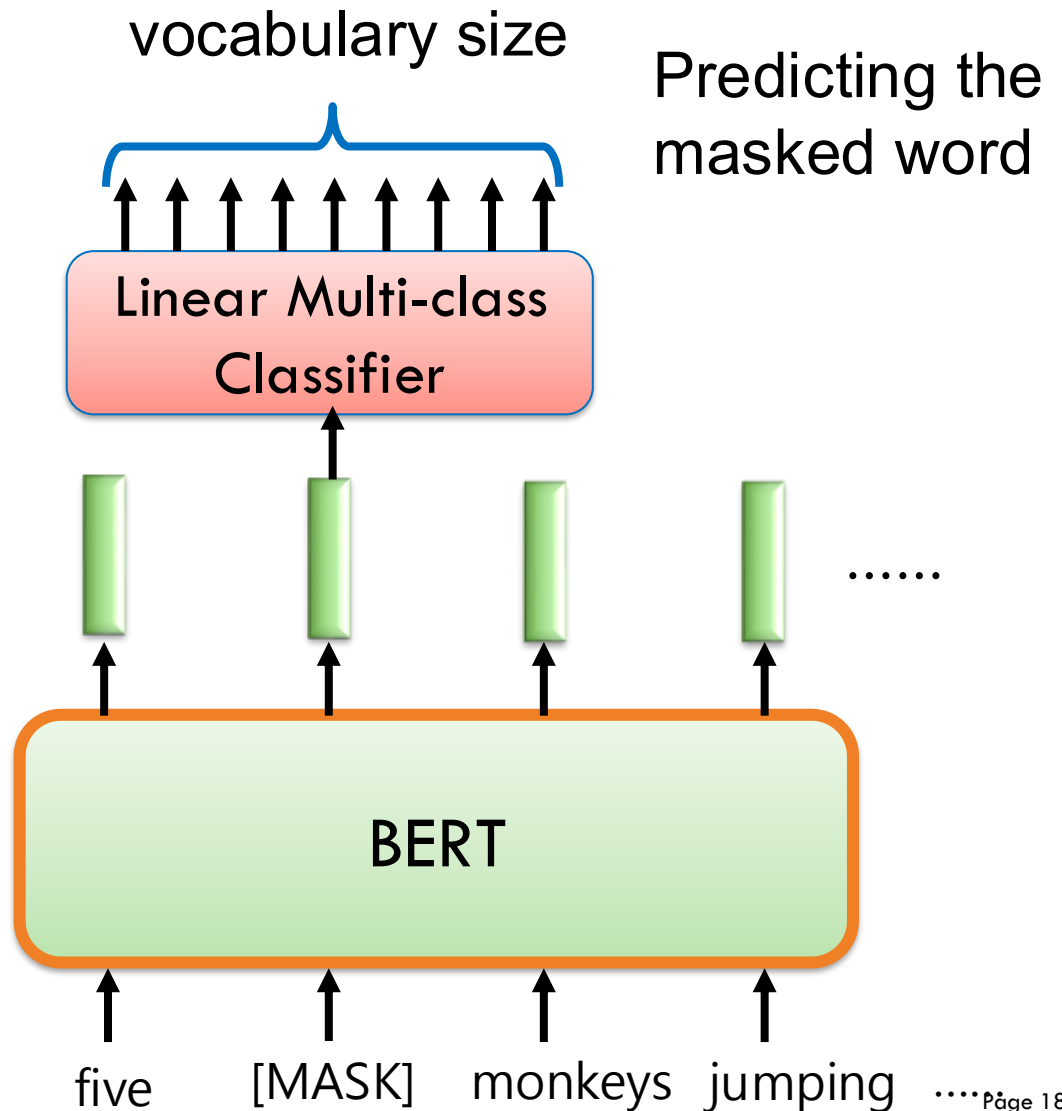


Encoder



Training of BERT

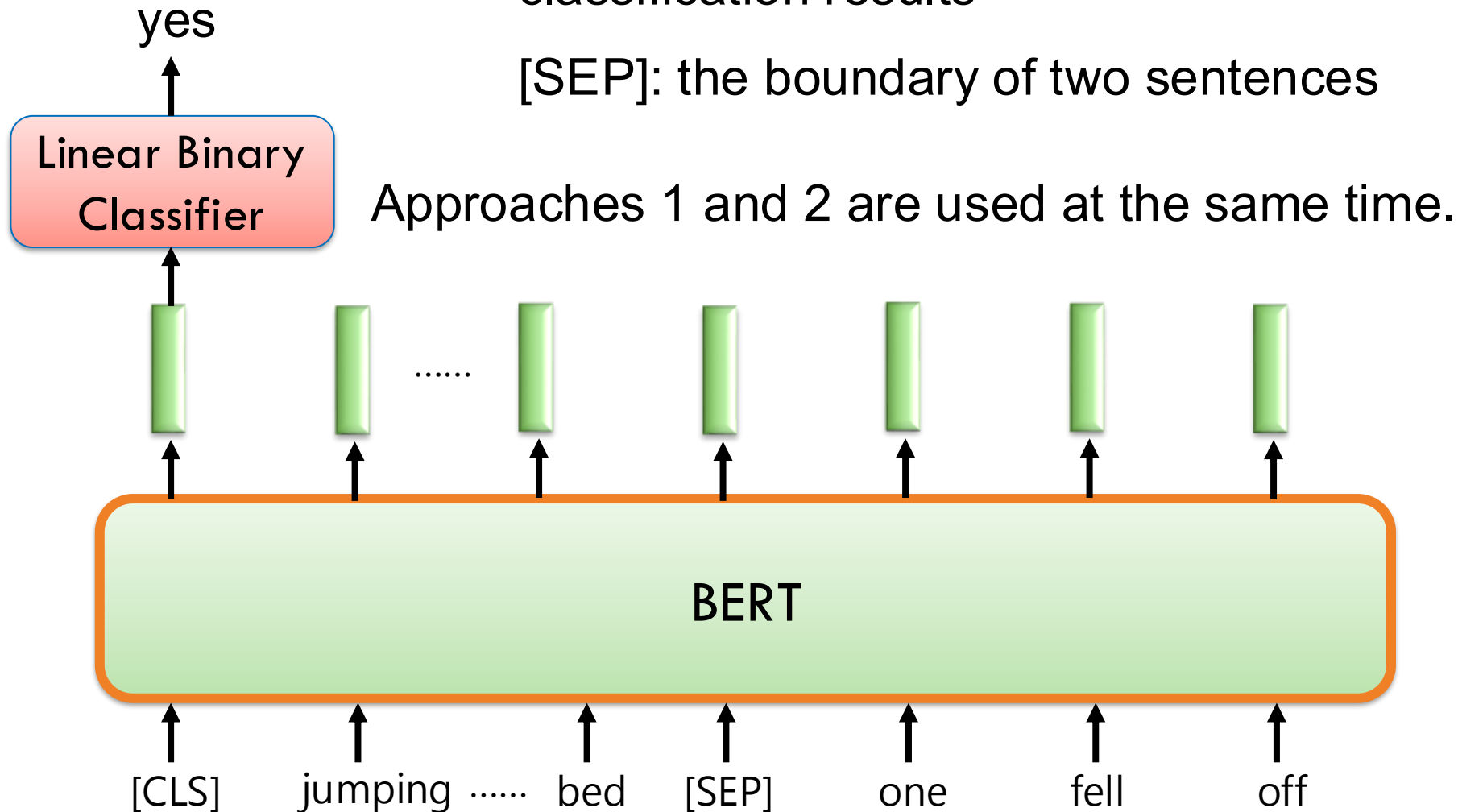
- Approach 1:
Masked LM



Training of BERT Approach 2: Next Sentence Prediction

[CLS]: the position that outputs classification results

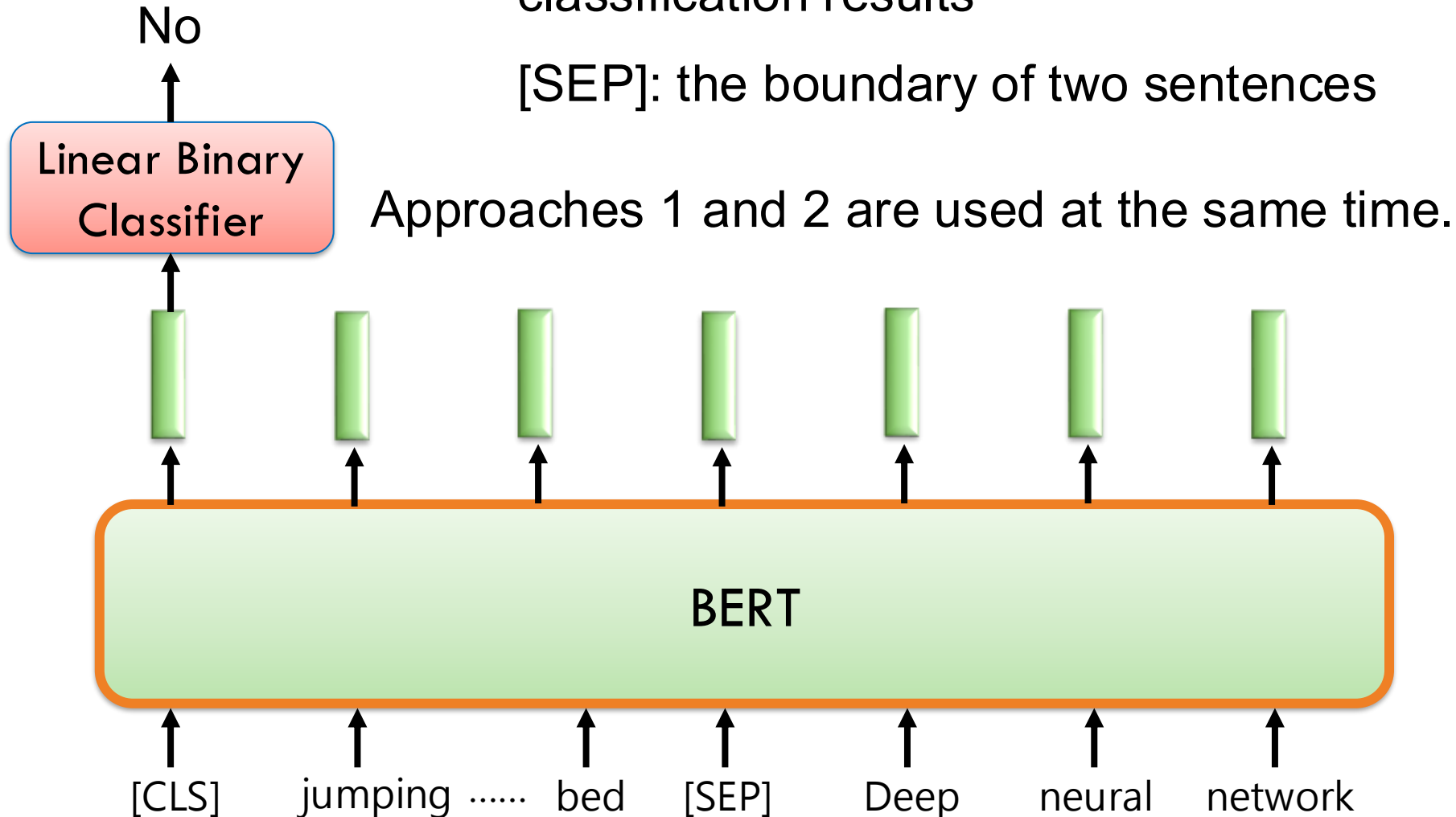
[SEP]: the boundary of two sentences



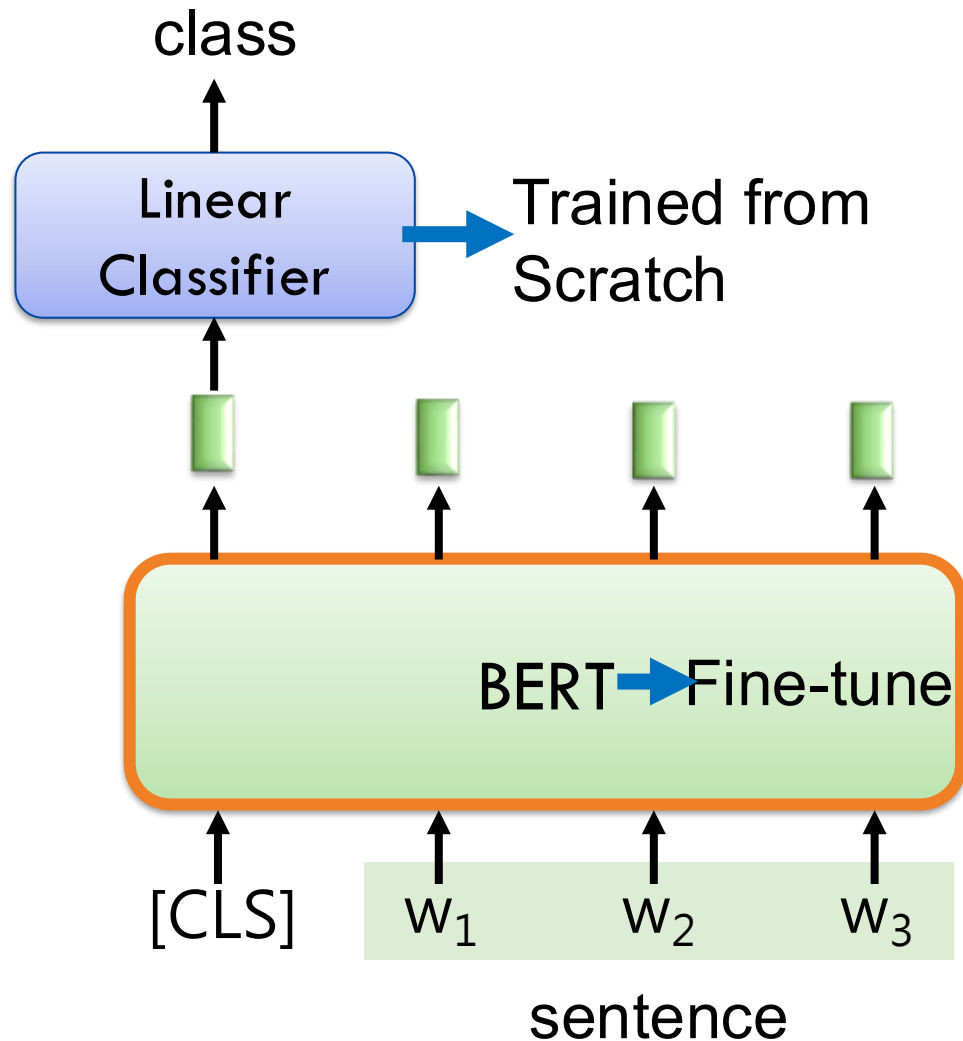
Training of BERT Approach 2: Next Sentence Prediction

[CLS]: the position that outputs classification results

[SEP]: the boundary of two sentences



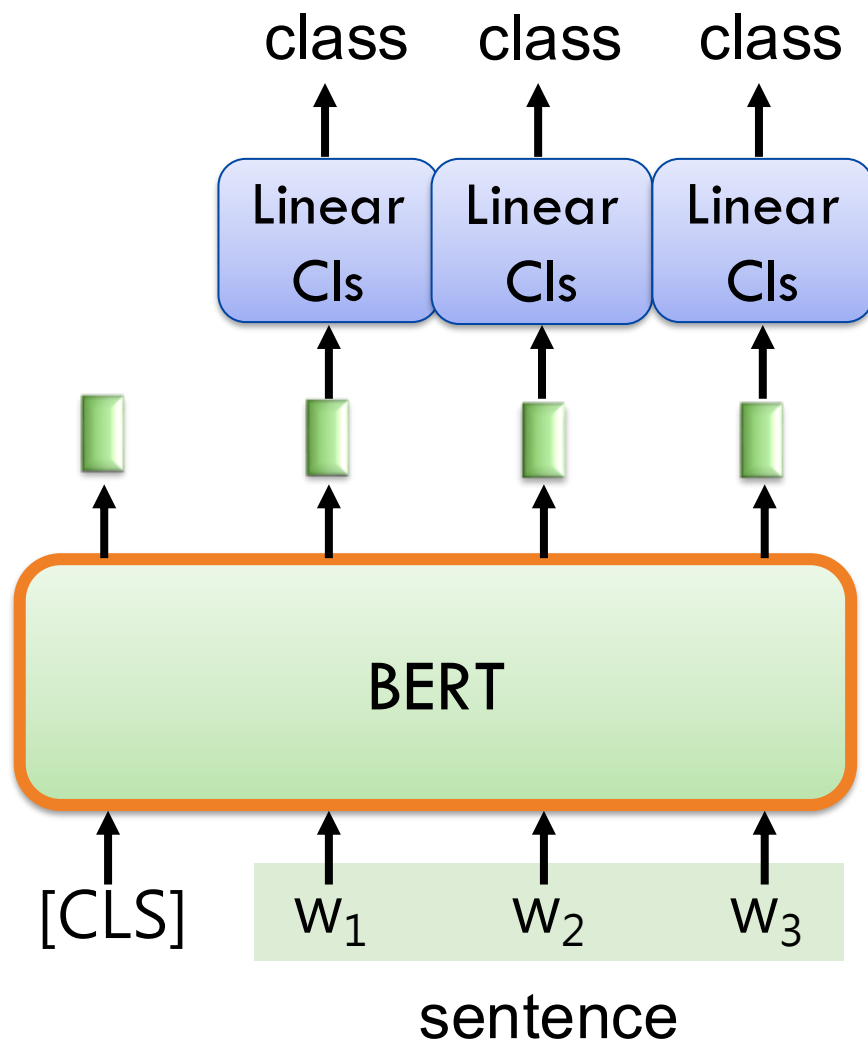
How to use BERT – Case 1



Input: single sentence,
output: class

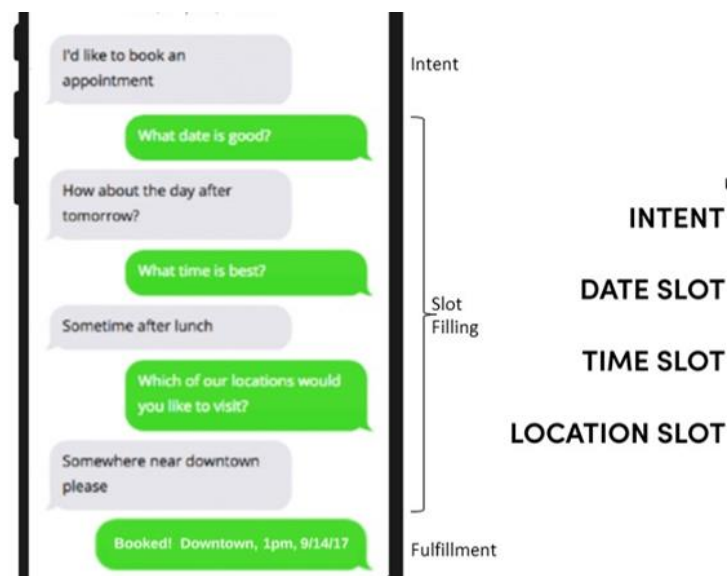
Example:
Sentiment analysis
Document Classification

How to use BERT – Case 2



Input: single sentence,
output: class of each word

Example: Slot filling

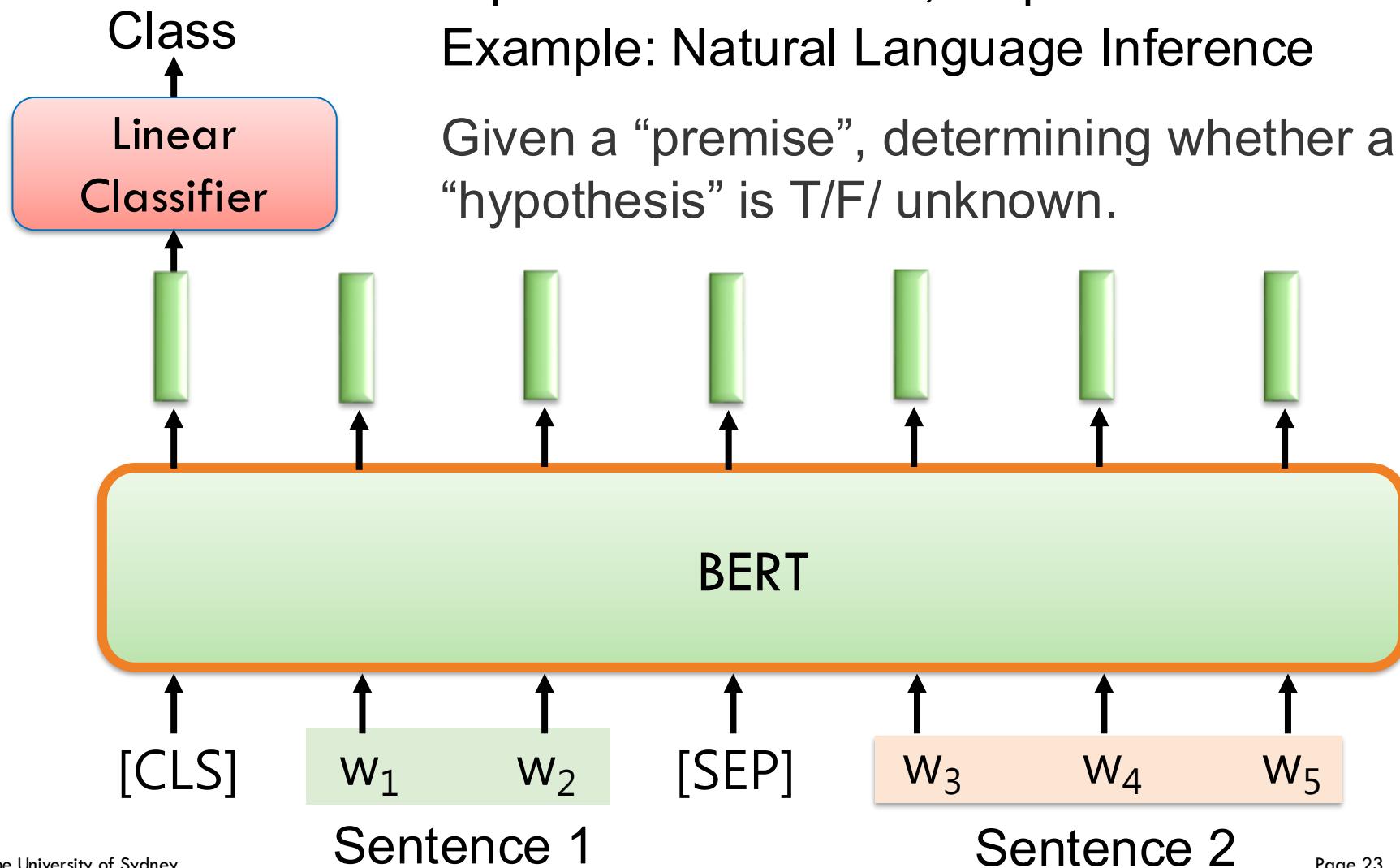


How to use BERT – Case 3

Input: two sentences, output: class

Example: Natural Language Inference

Given a “premise”, determining whether a “hypothesis” is T/F/ unknown.

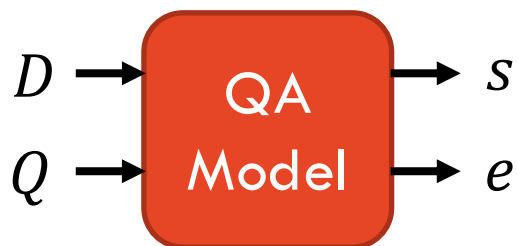


How to use BERT – Case 4

- Extraction-based Question Answering (QA) (E.g. SQuAD)

Document: $D = \{d_1, d_2, \dots, d_N\}$

Query: $Q = \{q_1, q_2, \dots, q_N\}$



output: two integers (s, e)

Answer: $A = \{q_s, \dots, q_e\}$

In meteorology, precipitation is any product of the condensation of 17 spheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **grau-pel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain 77 at 79 cations are called “showers”.

What causes precipitation to fall?

gravity

$s = 17, e = 17$

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?

grau-pel

Where do water droplets collide with ice crystals to form precipitation?

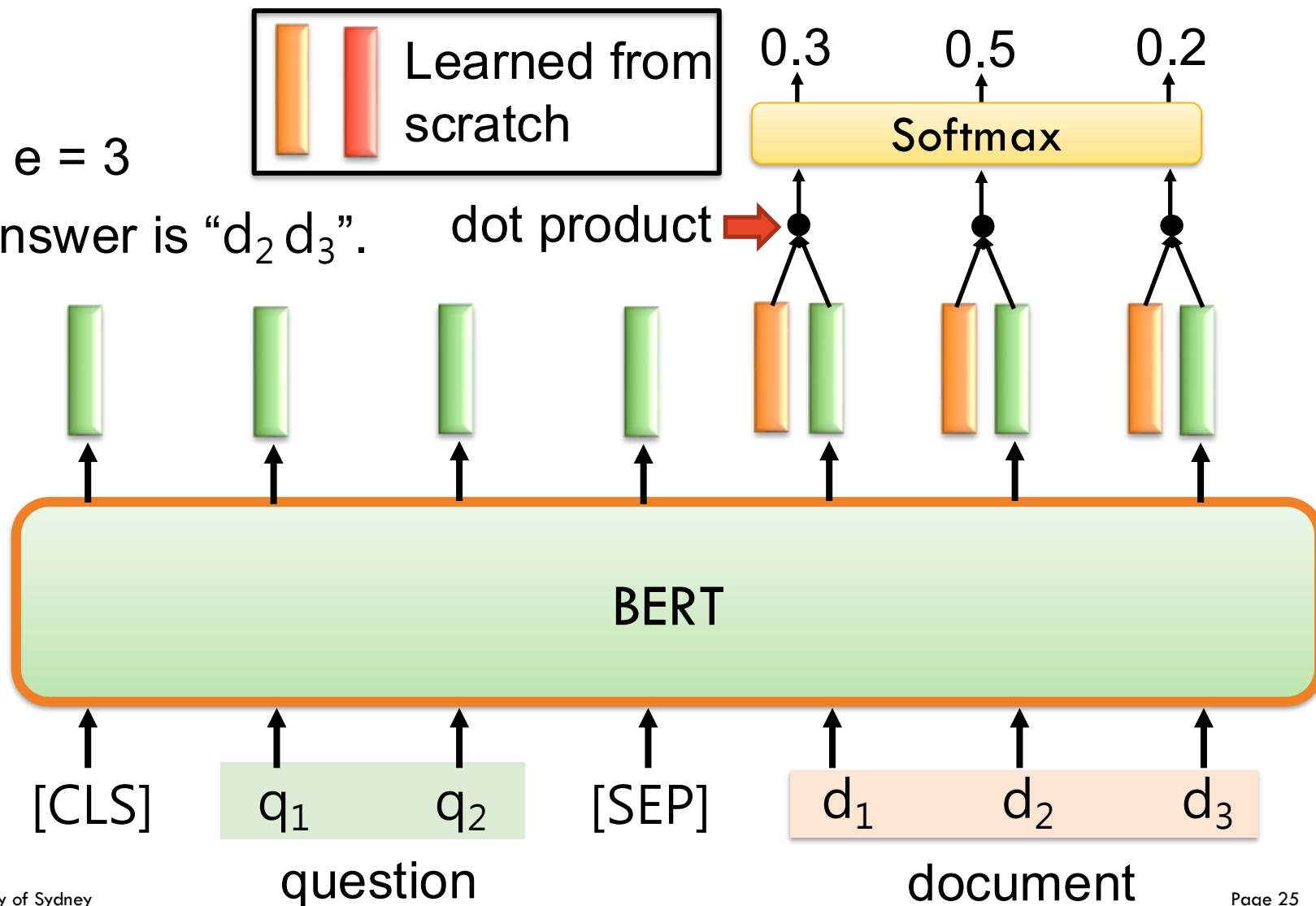
within a cloud

$s = 77, e = 79$

How to use BERT – Case 4

$s = 2, e = 3$

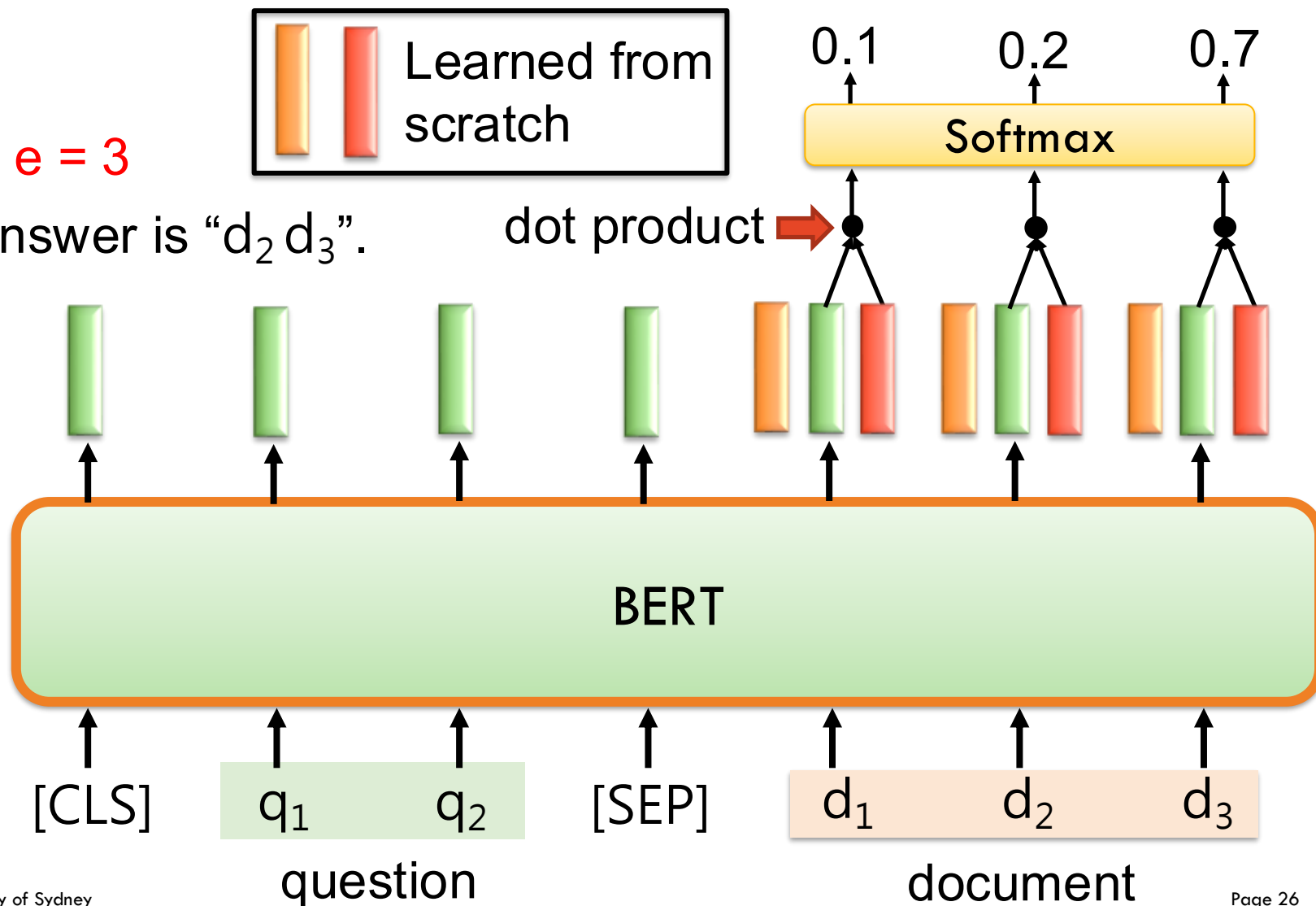
The answer is “ $d_2 d_3$ ”.



How to use BERT – Case 4

$s = 2$, $e = 3$

The answer is “ $d_2 d_3$ ”.

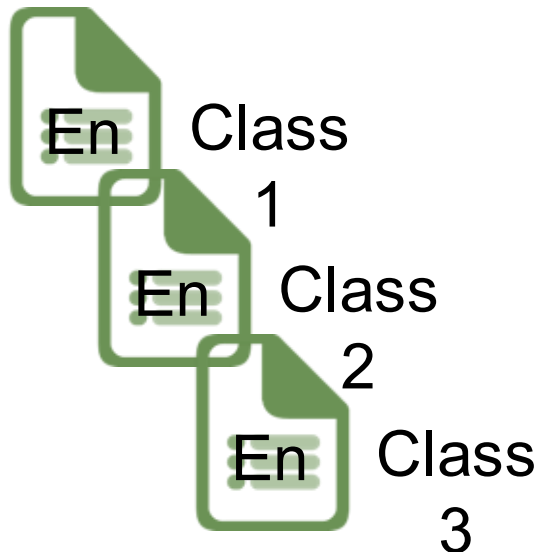


Multilingual BERT

Trained on 104 languages



Task specific training data for English



Task specific testing data for Chinese



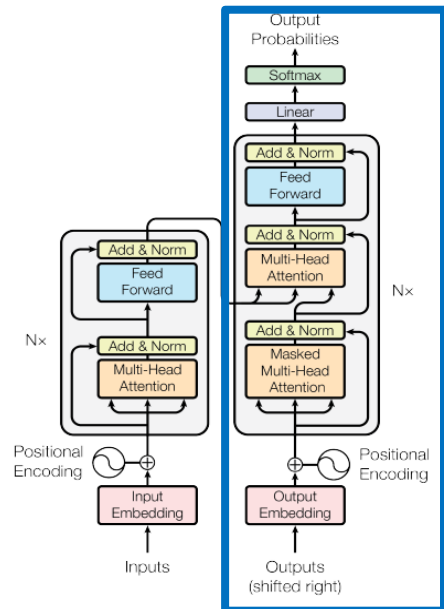
GPT

A **transformer** uses Encoder stack to model input, and uses Decoder stack to model output (using input information from encoder side).

If we are only interested in training a language model for the input for some other tasks, then we do not need the Decoder of the transformer, that gives us **BERT**.

But if we do not have input, we just want to model the “next word”, we can get rid of the Encoder side of a transformer and output “next word” one by one. This gives us **GPT**.

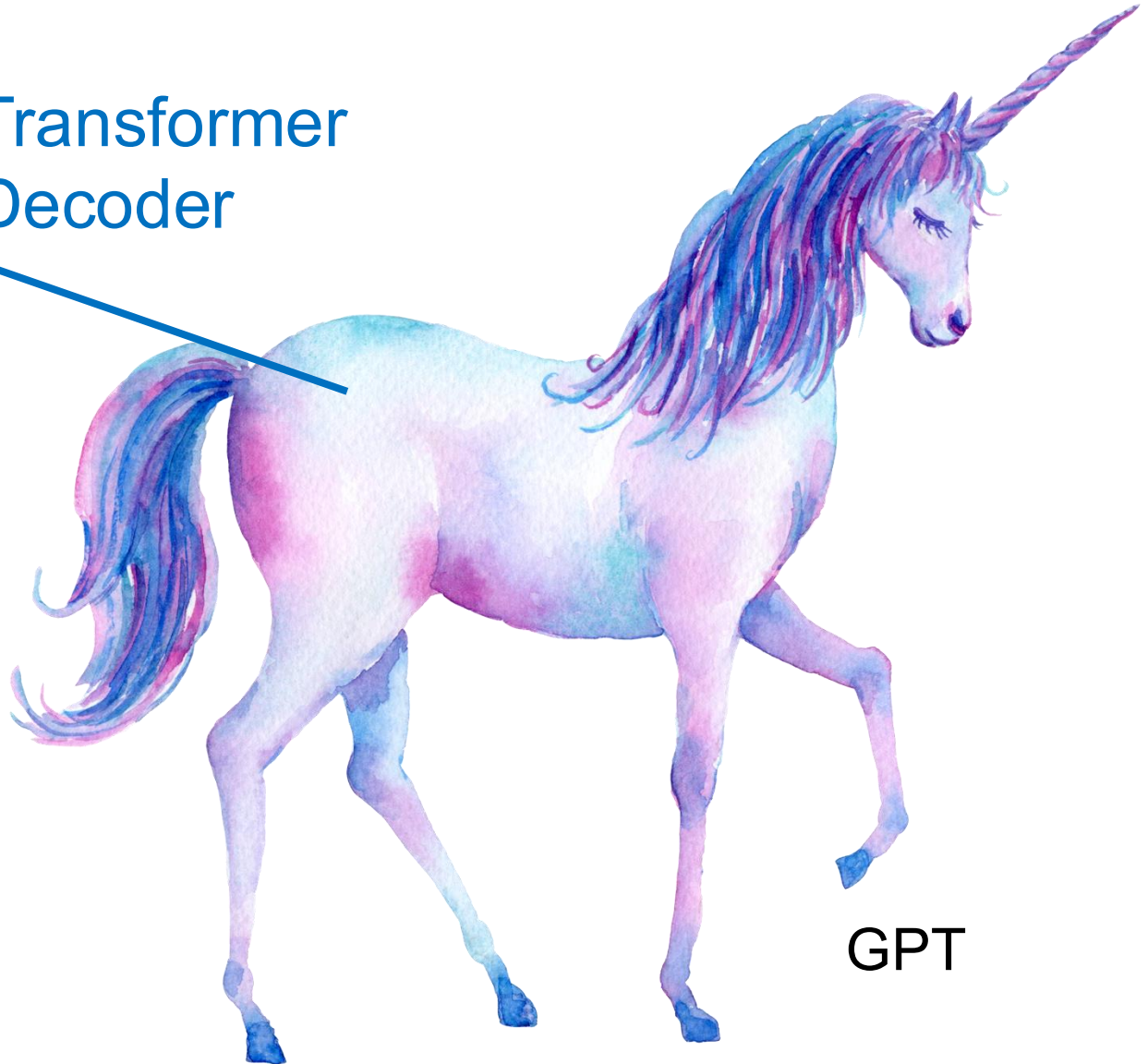
Generative Pre-Training (GPT)



Transformer
Decoder

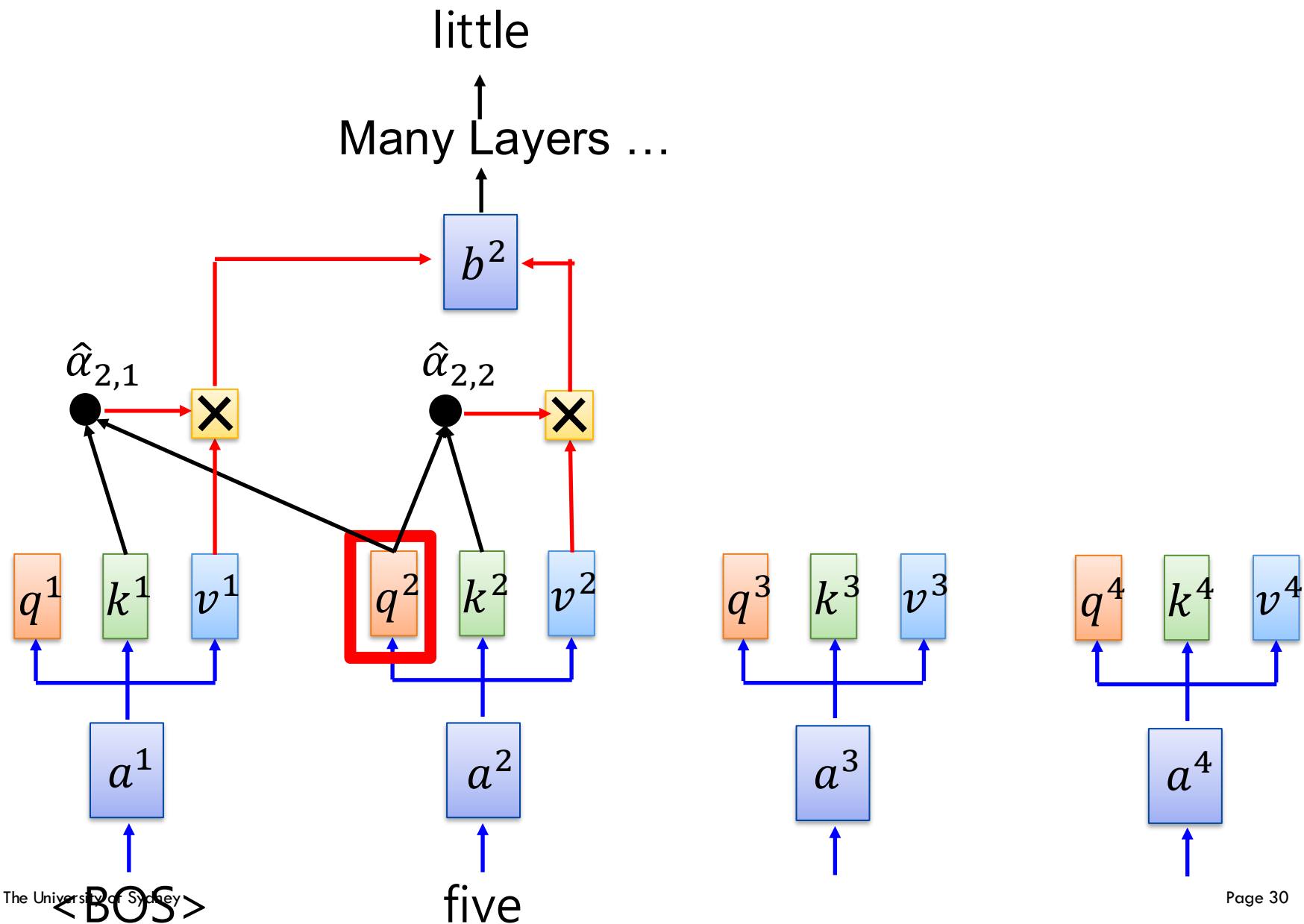
BERT
(340M)

ELMO
(94M)

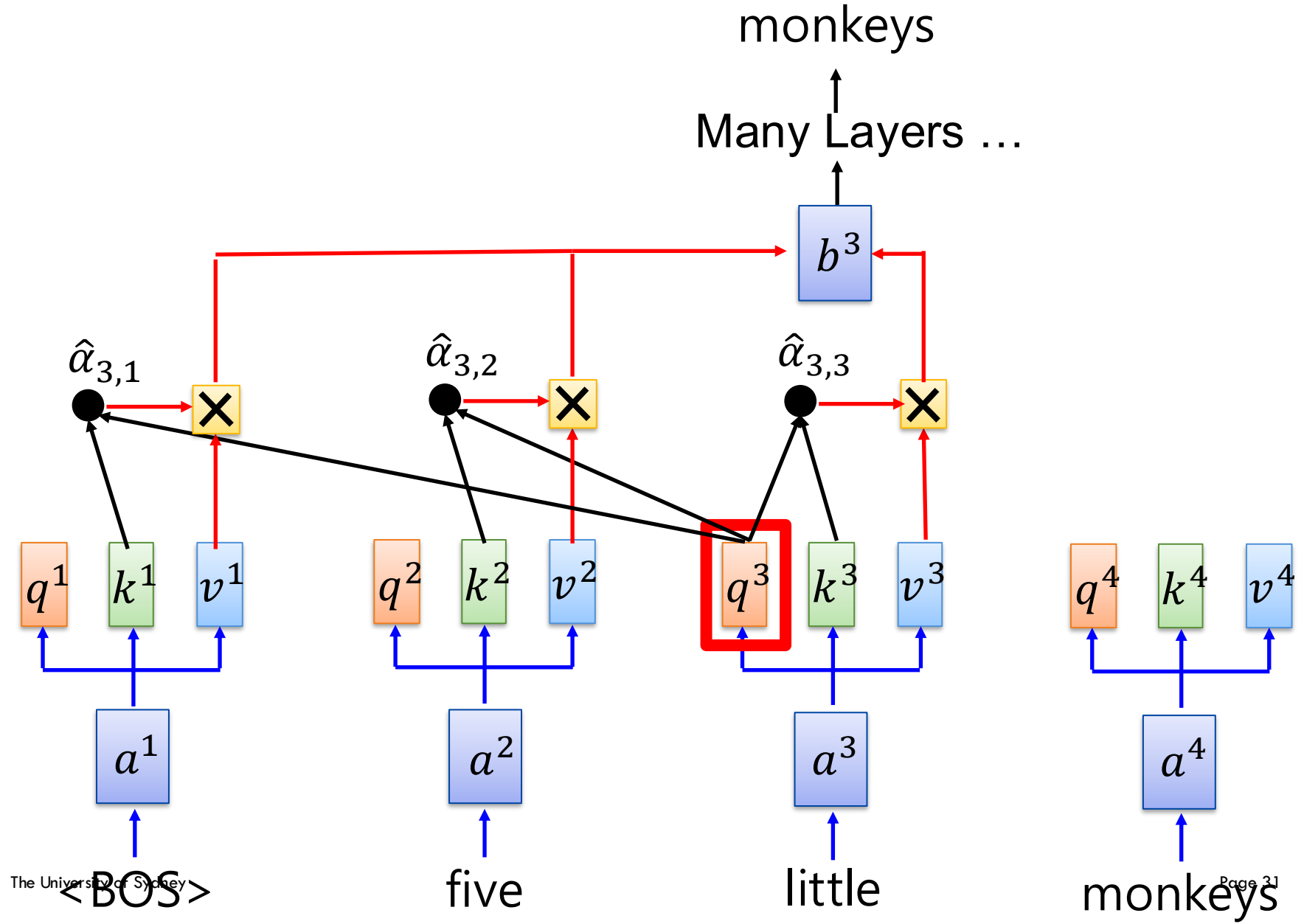


GPT

Generative Pre-Training (GPT)



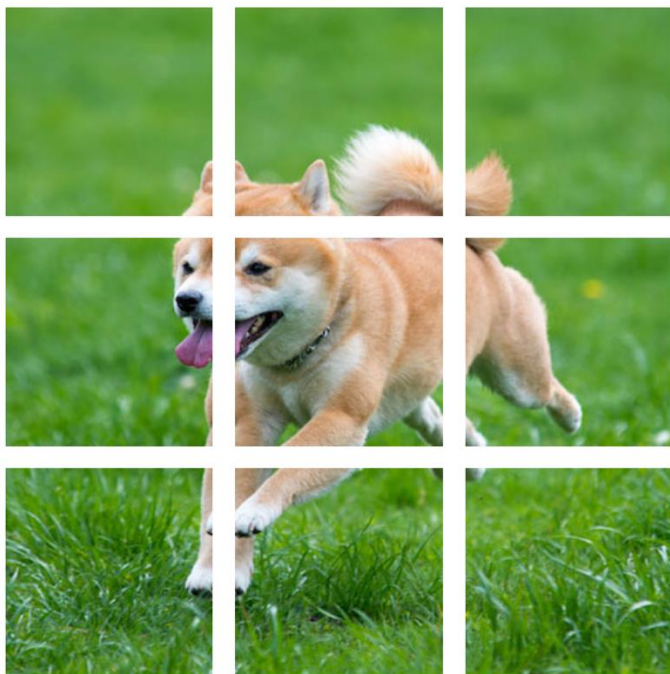
Generative Pre-Training (GPT)



Vision Transformer

How to represent an image with sequence?

1. Split an image into several multiple patches

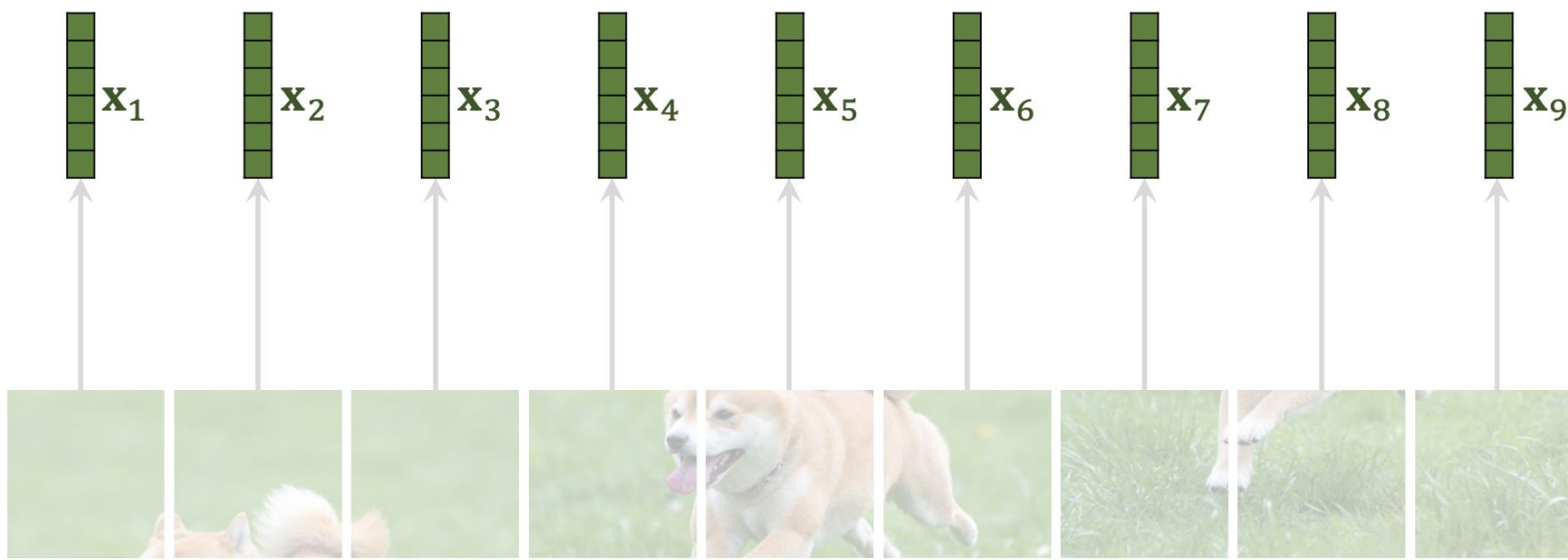


- Here, the patches do not overlap.
- The patches can overlap.
- User specifies:
 - **patch size**, e.g., 16×16 ;
 - **stride**, e.g., 16×16 .

How to represent an image with sequence?

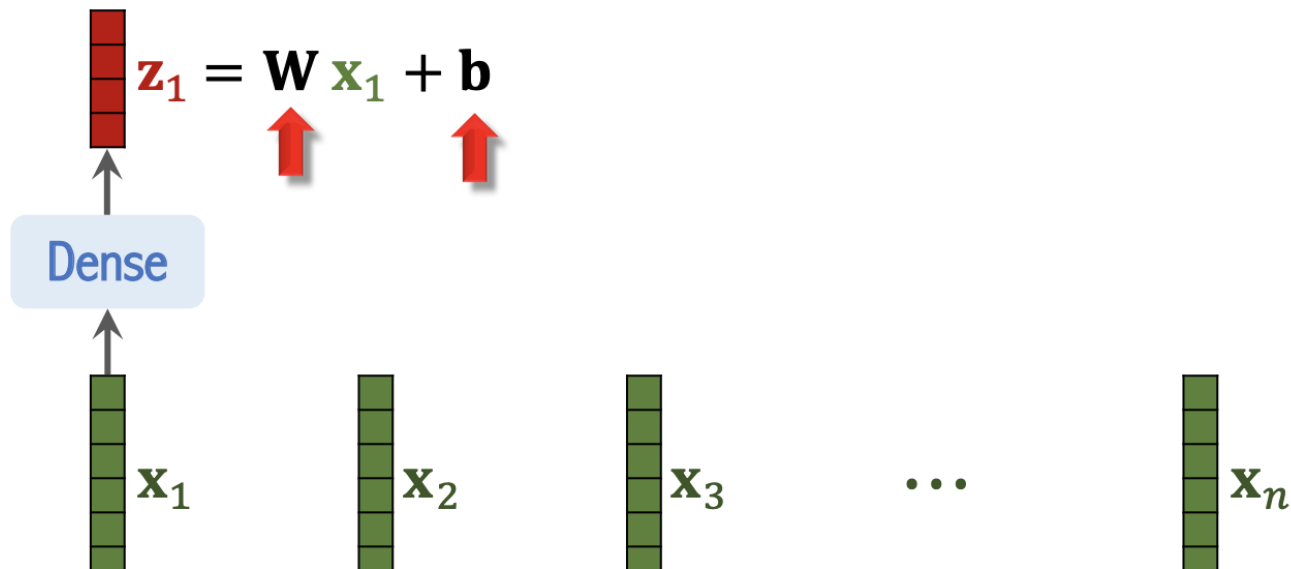
2. Vectorization

If the patches are $d_1 \times d_2 \times d_3$ tensors, then the vectors are $d_1 d_2 d_3 \times 1$.



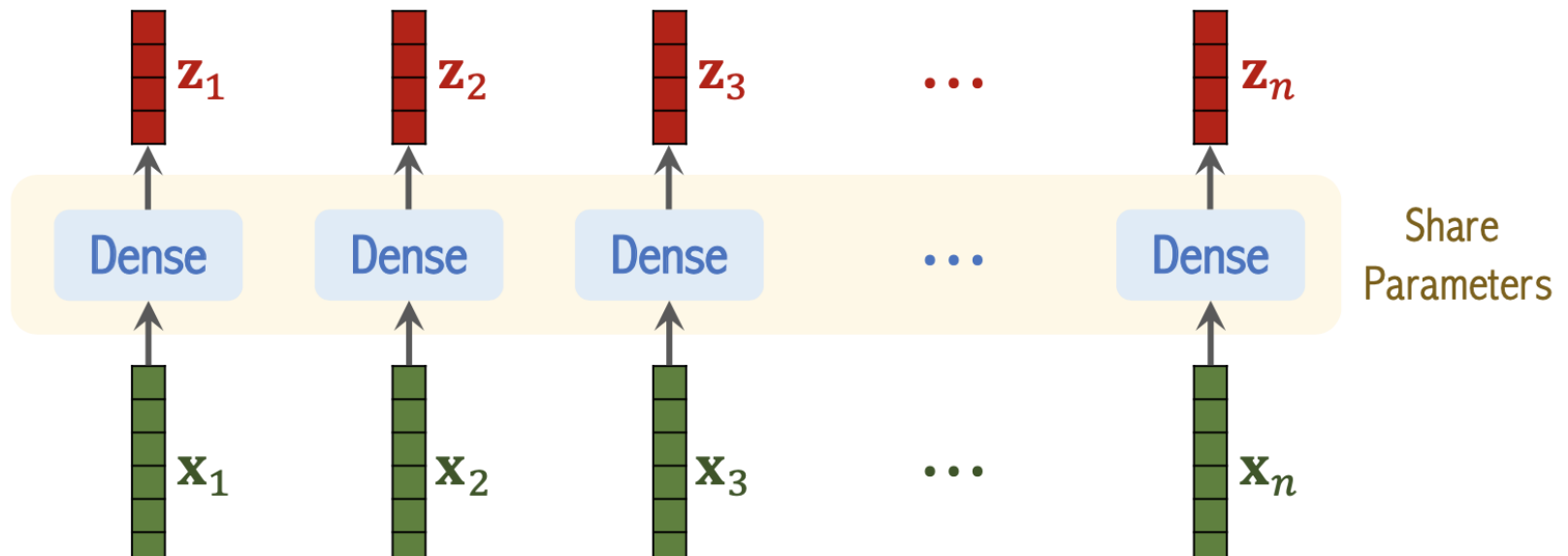
How to represent an image a with sequence?

2. Vectorization



How to represent an image a with sequence?

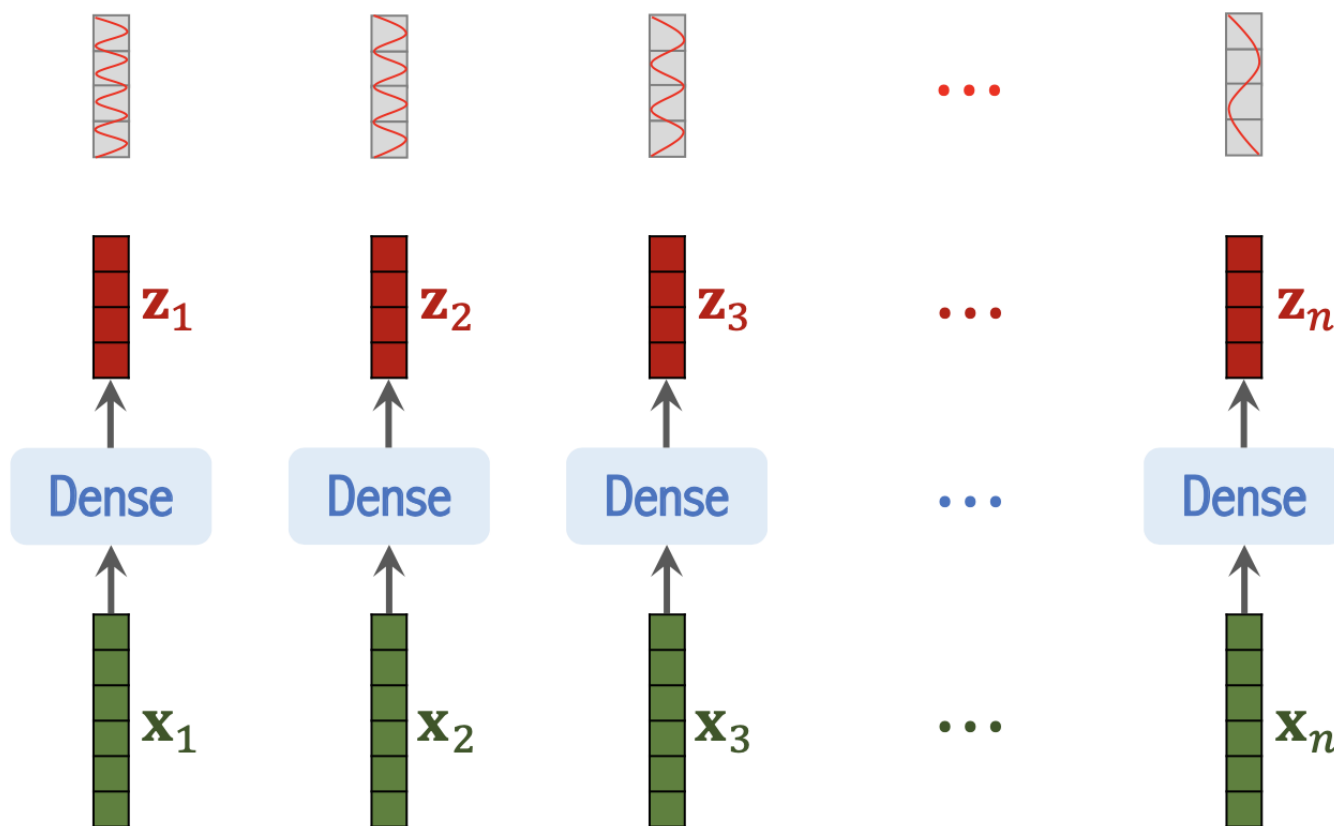
2. Vectorization



How to represent an image a with sequence?

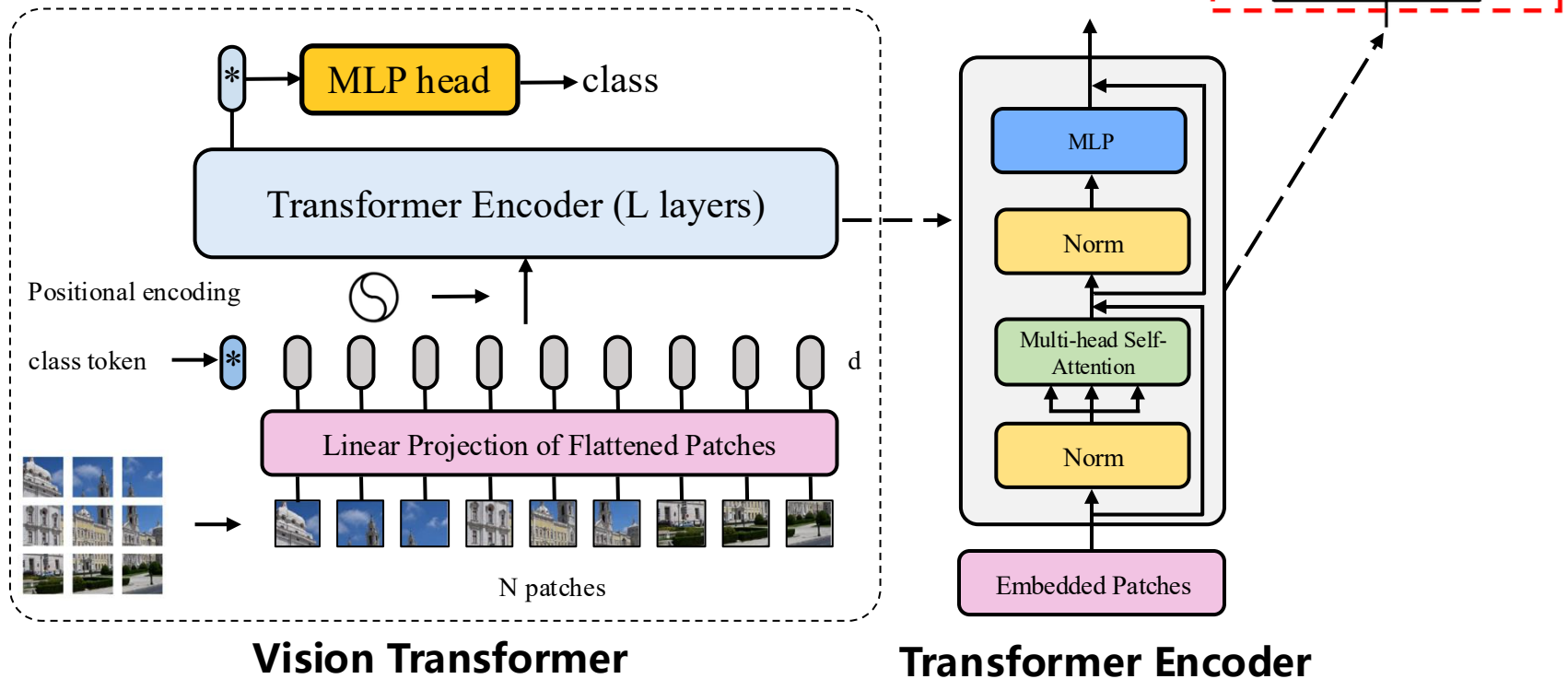
3. Add positional encoding

Add positional encoding vectors to $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_n$.



Vision Transformer

4. Overall Framework



Dosovitskiy et al., An Image is Worth 16*16 Words: Transformers for image recognition, ICLR, 2021, Google Brain

Vision Transformer

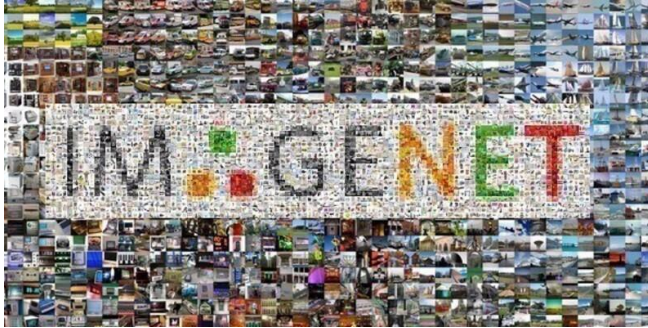
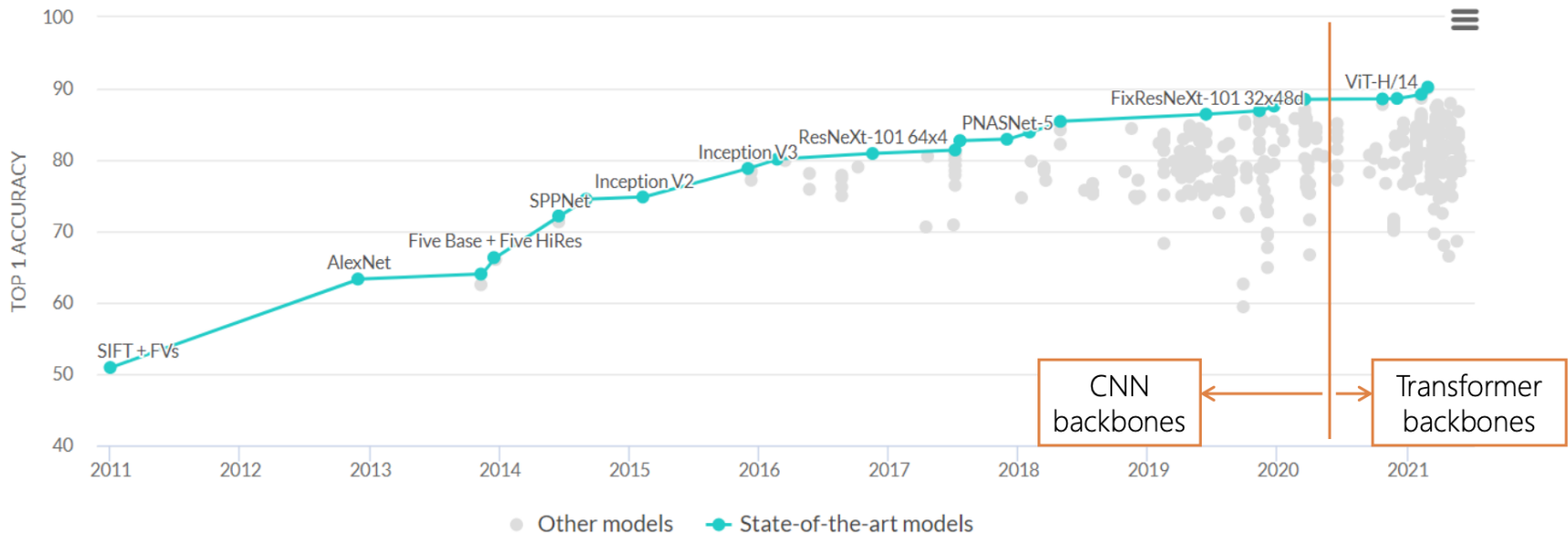


Image Classification Accuracy on ImageNet-1K

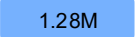
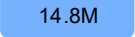
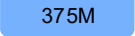
~1.2M Images, 1K Categories



<https://paperswithcode.com/sota/image-classification-on-imagenet>

Vision Transformer

ViT needs a large amount of training data to reach SOTA performance.

		ViT-B/16	ViT-B/32	ViT-L/16	ViT-L/32	ViT-H/14
ImageNet   	CIFAR-10	98.13	97.77	97.86	97.94	-
	CIFAR-100	87.13	86.31	86.35	87.07	-
	ImageNet	77.91	73.38	76.53	71.16	-
	ImageNet ReaL	83.57	79.56	82.19	77.83	-
	Oxford Flowers-102	89.49	85.43	89.66	86.36	-
	Oxford-IIIT-Pets	93.81	92.04	93.64	91.35	-
ImageNet-21k   	CIFAR-10	98.95	98.79	99.16	99.13	99.27
	CIFAR-100	91.67	91.97	93.44	93.04	93.82
	ImageNet	83.97	81.28	85.15	80.99	85.13
	ImageNet ReaL	88.35	86.63	88.40	85.65	88.70
	Oxford Flowers-102	99.38	99.11	99.61	99.19	99.51
	Oxford-IIIT-Pets	94.43	93.02	94.73	93.09	94.82
JFT-300M   	CIFAR-10	99.00	98.61	99.38	99.19	99.50
	CIFAR-100	91.87	90.49	94.04	92.52	94.55
	ImageNet	84.15	80.73	87.12	84.37	88.04
	ImageNet ReaL	88.85	86.27	89.99	88.28	90.33
	Oxford Flowers-102	99.56	99.27	99.56	99.45	99.68
	Oxford-IIIT-Pets	95.80	93.40	97.11	95.83	97.56

EfficientNet-B7
Top-1 Acc. 84.3%

Vision Transformer

Improved Variants

🔗 Swin Transformer

📊 Ranked #3 Object Detection on COCO test-dev (using additional training data)

📊 Ranked #2 Instance Segmentation on COCO test-dev (using additional training data)

📊 Ranked #2 Object Detection on COCO minival (using additional training data)

📊 State of the Art Instance Segmentation on COCO minival (using additional training data)

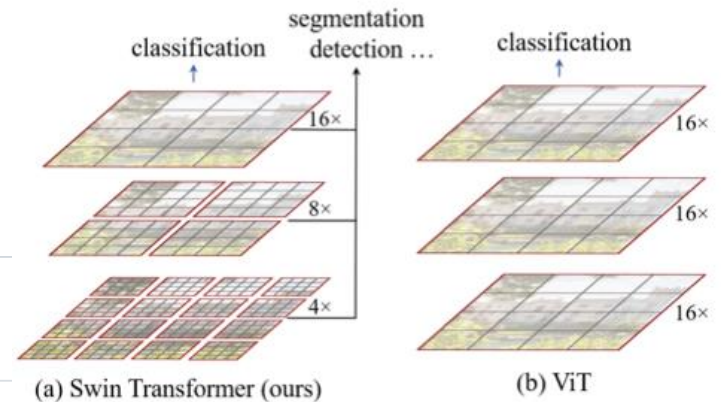
📊 Ranked #11 Semantic Segmentation on ADE20K (using additional training data)

📊 Ranked #11 Semantic Segmentation on ADE20K val

📊 State of the Art Action Recognition on Something-Something V2 (using additional training data)

📊 Ranked #3 Action Classification on Kinetics-400 (using additional training data)

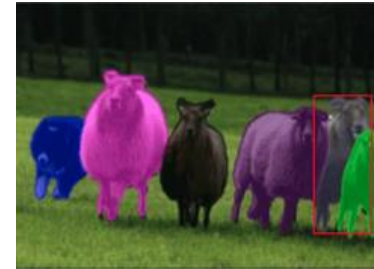
📊 Ranked #2 Action Classification on Kinetics-600 (using additional training data)



Object Detection



Semantic Segmentation



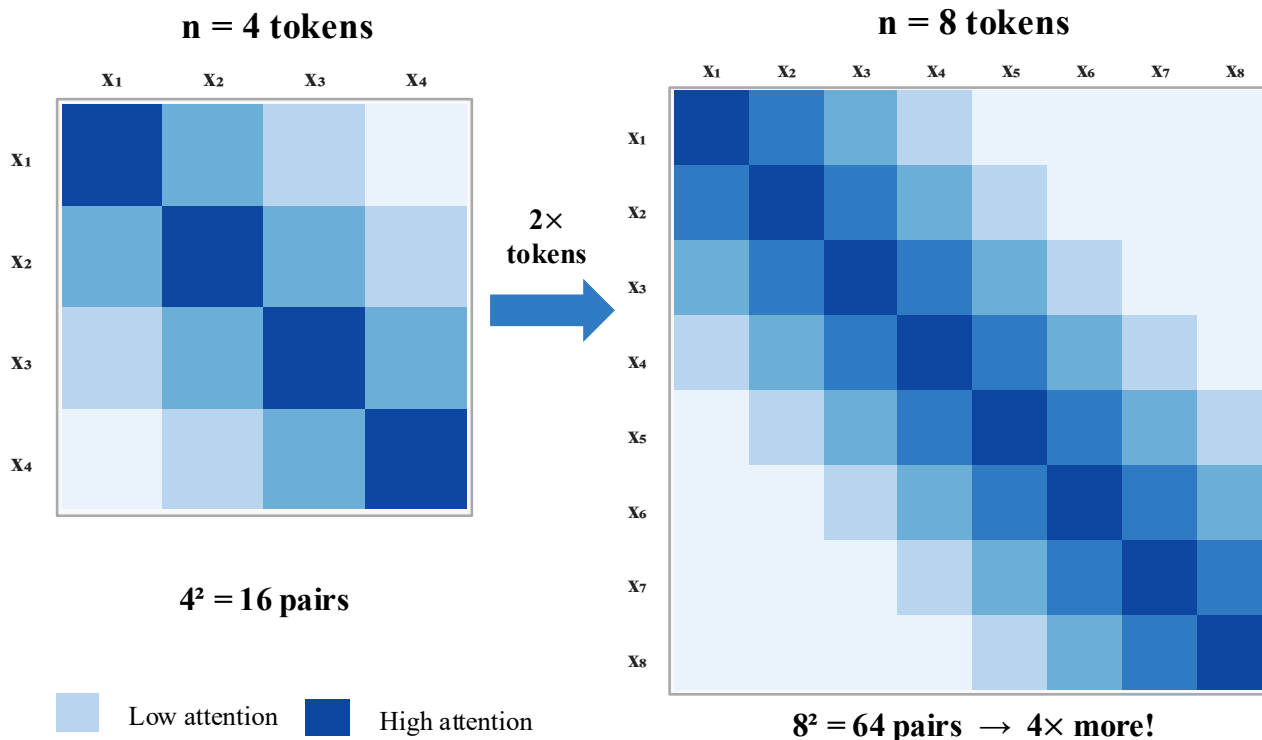
Instance Segmentation

Liu et al., Swin Transformer: Hierarchical Vision Transformer using Shifted Windows, ICCV 2021 (**Best Paper**).

Mamba – A new S2S Architecture

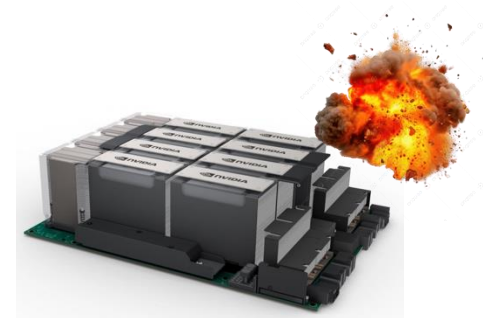
Why Transformer Struggles with Long Sequences?

- ❑ Self-attention computes a score between every pair of tokens:
cost grows as $O(n^2)$
- ❑ Doubling the sequence length quadruples the attention matrix — **memory & compute explode**

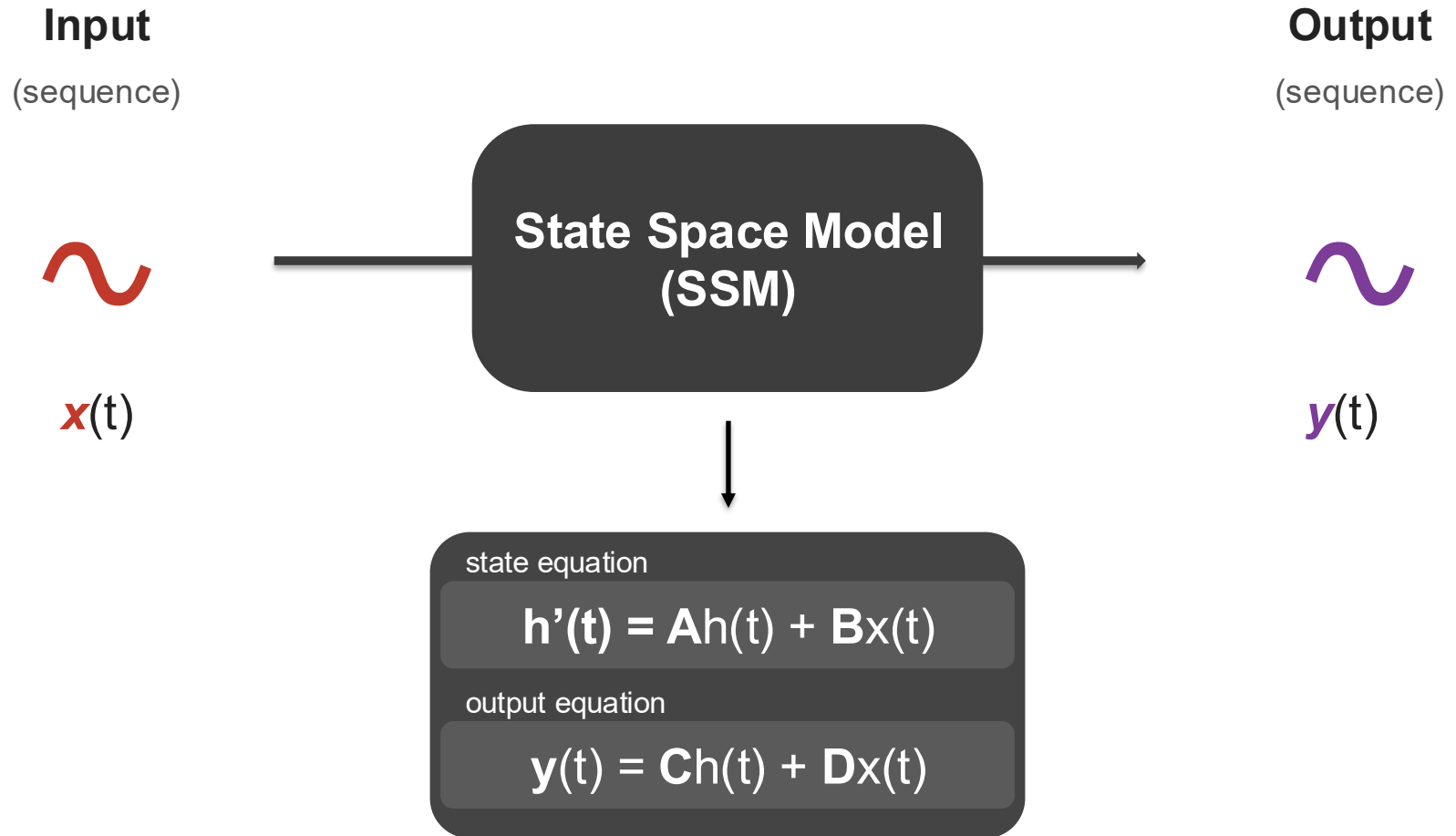


Context Window :

- GPT-3: 4K
- GPT-4: 32K
- GPT-5: ~200K

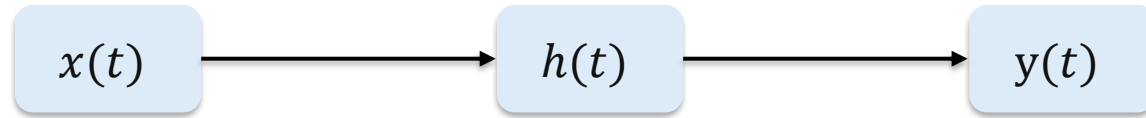


State Space Models



R. E. Kalman. "A New Approach to Linear Filtering and Prediction Problems." ASME 1960

Computing with SSM: Recurrent View

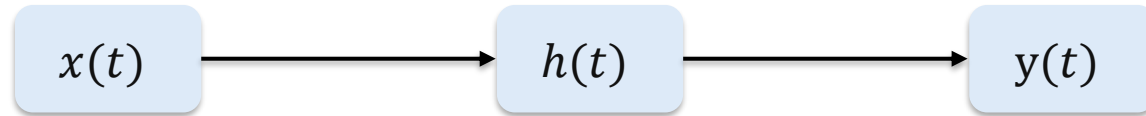


$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \qquad y(t) = \mathbf{C}h(t) + \mathbf{D}x(t)$$

Euler Discretization:

$$\begin{aligned} h(t + \Delta) &\approx h(t) + \Delta h'(t) && \text{(First-order approximation)} \\ &= h(t) + \Delta(\mathbf{A}h(t) + \mathbf{B}x(t)) \\ &= (\mathbf{I} + \Delta\mathbf{A})h(t) + \Delta\mathbf{B}x(t) \\ &= \bar{\mathbf{A}}h(t) + \bar{\mathbf{B}}x(t) \end{aligned}$$

Computing with SSM: Recurrent View



$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t) \qquad y(t) = \mathbf{C}h(t) + \mathbf{D}x(t)$$

1. Discretize

$$\bar{\mathbf{A}} = (\mathbf{I} + \Delta\mathbf{A}) \qquad \bar{\mathbf{B}} = \Delta\mathbf{B}$$

2. Recurrent “hidden state”

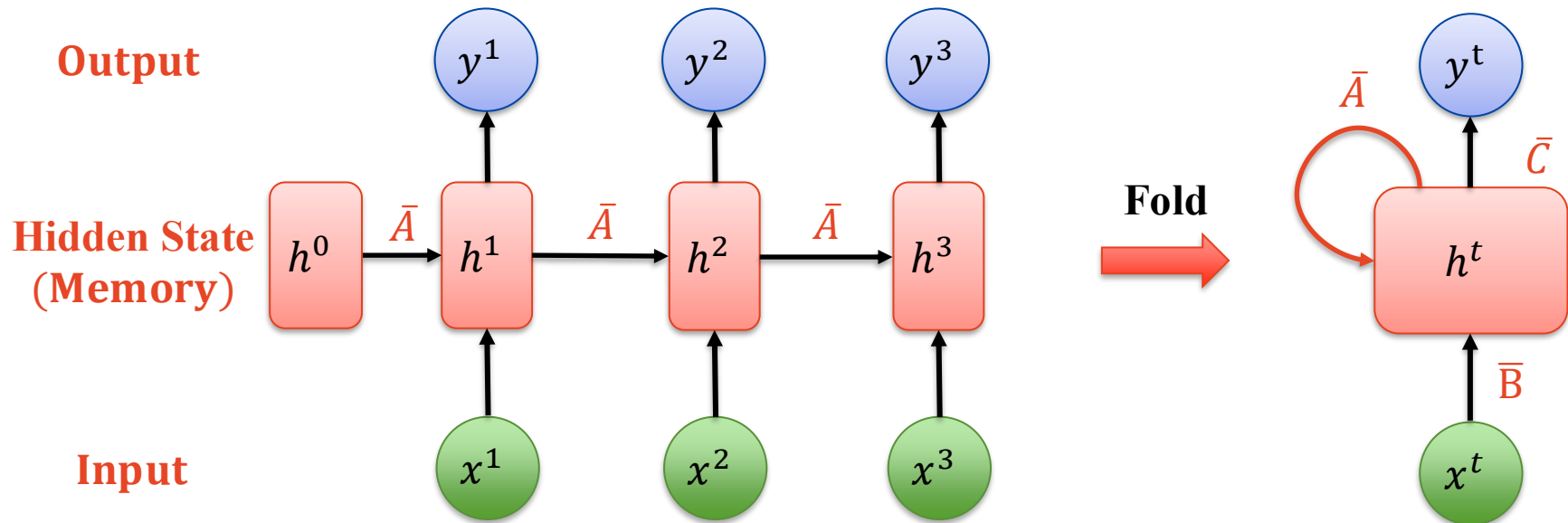
$$h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t$$

3. Out projection

$$y_t = \bar{\mathbf{C}}h_{t-1} + \bar{\mathbf{D}}x_t$$

Can be unrolled as a linear RNN

Recurrent Computation

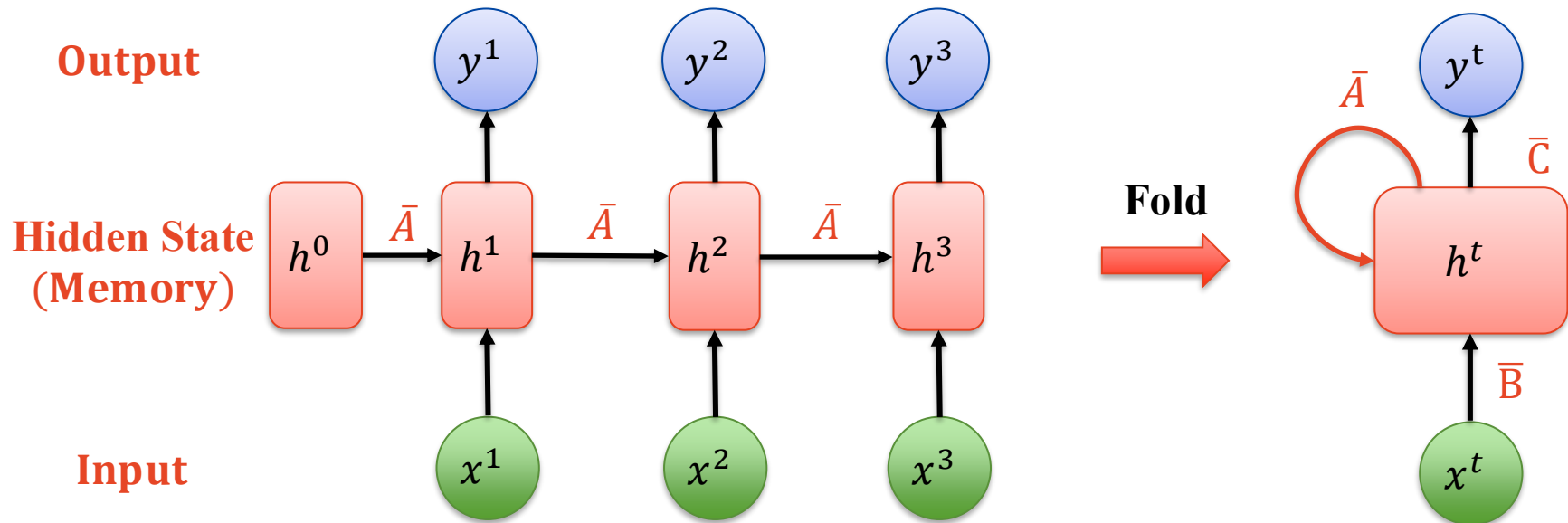


- Infinite Context Window
- Complexity: $O(N)$
- Not **parallelizable**

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t$$

$$y_t = \bar{C}h_{t-1} + \bar{D}x_t$$

Recurrent Computation



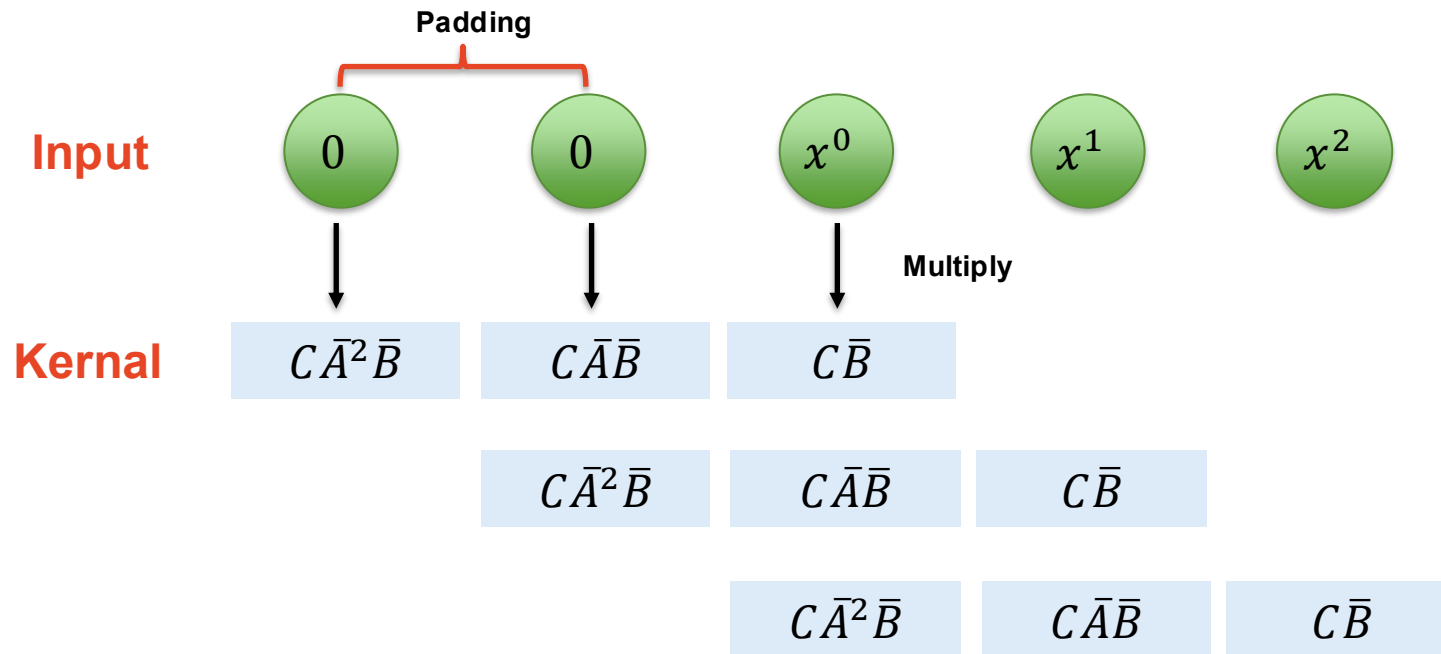
$$h_1 = \bar{A}h_0 + \bar{B}x_0$$

$$h_2 = \bar{A}h_1 + \bar{B}x_1 = \bar{A}(\bar{A}h_0 + \bar{B}x_0) + \bar{B}x_1 = \bar{A}^2h_0 + \bar{A}\bar{B}x_0 + \bar{B}x_1$$

⋮

$$h_t = \bar{A}^t h_0 + \sum_{t=0}^L \bar{A}^{L-t} \bar{B} x_t \quad \longrightarrow \quad y_t = \sum_{t=0}^L \bar{C} \bar{A}^{L-t} \bar{B} x_t + D x_t$$

Represent SSMs with Convolution



Output

$$y_0 = C\bar{B}x_0$$

$$y_1 = C\bar{A}\bar{B}x_0 + C\bar{B}x_1$$

$$y_2 = C\bar{A}^2\bar{B}x_0 + C\bar{A}\bar{B}x_1 + C\bar{B}x_2$$

$$K = (C\bar{A}^2\bar{B}, C\bar{A}\bar{B}, C\bar{B})$$

$$Y = K * X$$

$$y^0$$

$$y^1$$

$$y^2$$

Training and Inference of SSMs

Training Mode

- Complexity: $O(n)$
- Parallelizable

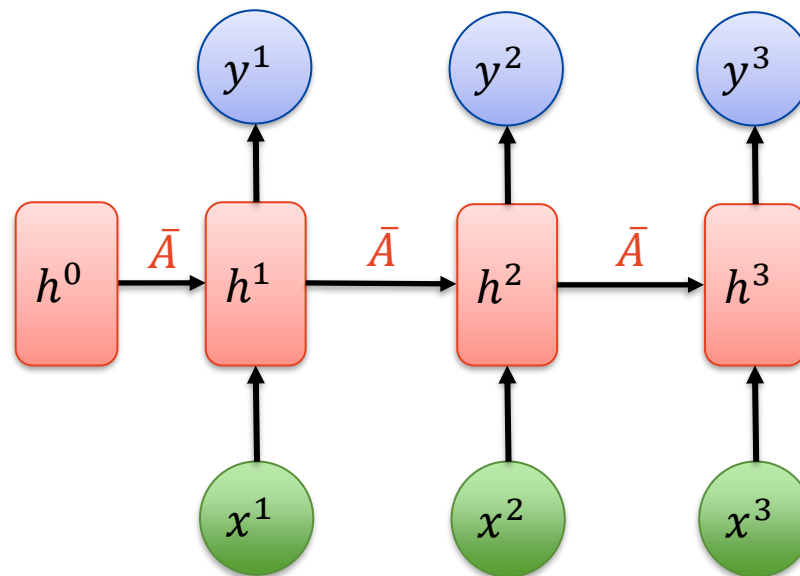
$$K \in \mathbb{R}^L = (C\bar{A}^{L-1}\bar{B}, C\bar{A}^{L-2}\bar{B}, \dots, C\bar{A}\bar{B}, C\bar{B})$$

$$Y = K * X$$

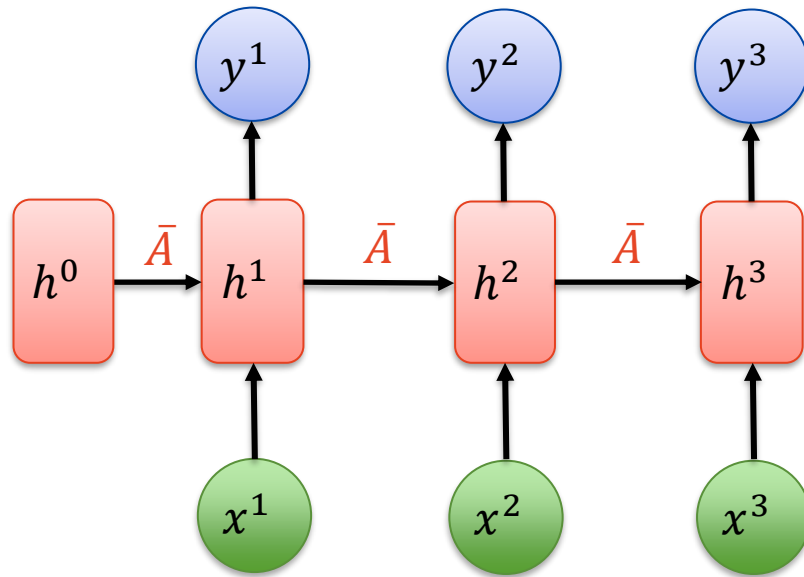
Can be computed with convolutions, similar to CNN

Inference Mode

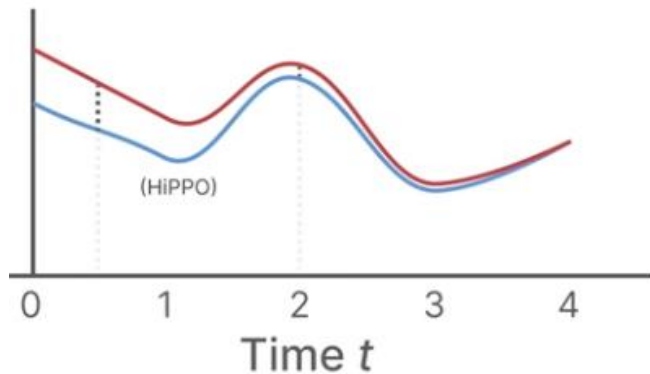
- Infinite Context
- Complexity: $O(1)$



The Importance of A Matrix



The **A** matrix captures information about the previous state to build the new state.



HiPpo

$$A_{nk} = \begin{cases} 0 & n < k \\ n + 1 & n = k \\ 2n + 1 & n > k \end{cases}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 3 & 3 & 0 \\ 1 & 3 & 5 & 4 \end{bmatrix}$$

With HiPpo, the model does not forget something at the beginning of a sequence.

Structured State Space Models (S4)

Initialize Parameters

A (with Hippo), B, C

Computer Kernel

K

Convolution

$y = x * K$

```
def parameters(N, dt_min=1e-3, dt_max=1e-1):
    # Initialization
    log_dt = np.random.rand() * (np.log(dt_max)-np.log(dt_min)) + np.log(dt_min)
    A = -0.5 + 1j * np.pi * np.arange(N) # S4D-Lin initialization
    B = np.ones(N)
    C = np.random.randn(N) + 1j * np.random.randn(N) # Variance preserving initia
    return log_dt, np.log(-A.real), A.imag, B, C

def kernel(L, log_dt, log_A_real, A_imag, B, C):
    # Discretization (e.g. bilinear transform)
    dt, A = np.exp(log_dt), -np.exp(log_a_real) + 1j * a_imag
    dA, dB = (1+dt*A/2) / (1-dt*A/2), dt*B / (1-dt*A/2)

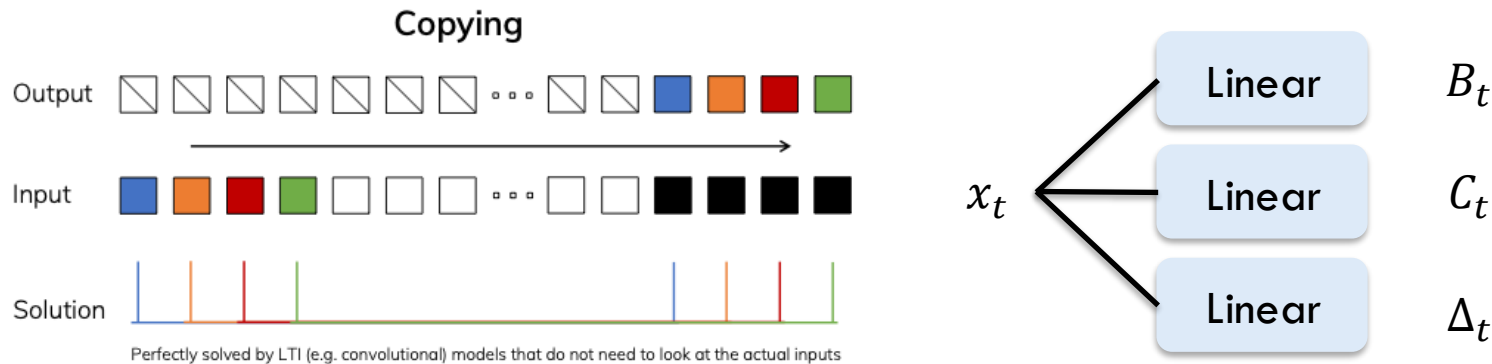
    # Computation (Vandermonde matrix multiplication - can be optimized)
    return (B*C) @ (dA[:, None] ** np.arange(L))

def forward(u, parameters):
    L = u.shape[-1]
    K = kernel(L, *parameters)
    K_f, u_f = np.fft.fft(K, n=2*L), np.fft.fft(input, n=2*L) # Convolve y = u
    return np.fft.ifft(K_f*u_f, n=2*L)[..., :L]
```

Gu et al., Efficiently Modeling Long Sequences with Structured State Spaces, ICLR 2022. (Outstanding Paper)

Mamba – Selective State Space Models (S6)

1. Improve SSMs with selection (input-dependency in matrices)

**Algorithm 1** SSM (S4)

Input: $x : (B, L, D)$

Output: $y : (B, L, D)$

1: $\mathbf{A} : (D, N) \leftarrow \text{Parameter}$

- Represents structured $N \times N$ matrix

2: $\mathbf{B} : (\mathcal{D}, \mathcal{N}) \leftarrow \text{Parameter}$ 3: $C : (D, N) \leftarrow \text{Parameter}$ 4: $\Delta : (D) \leftarrow \tau_{\Delta}(\text{Parameter})$ 5: $\overline{A}, \overline{B} : (D, N) \leftarrow \text{discretize}(\Delta, A, B)$ 6: $y \leftarrow \text{SSM}(\overline{A}, \overline{B}, C)(x)$

- Time-invariant: recurrence or convolution

7: **return** y **Algorithm 2** SSM + Selection (S6)

Input: $x : (B, L, D)$

Output: $y : (B, L, D)$

1: $\mathbf{A} : (D, N) \leftarrow \text{Parameter}$

- Represents structured $N \times N$ matrix

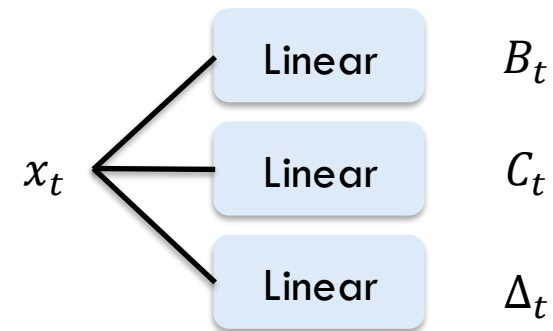
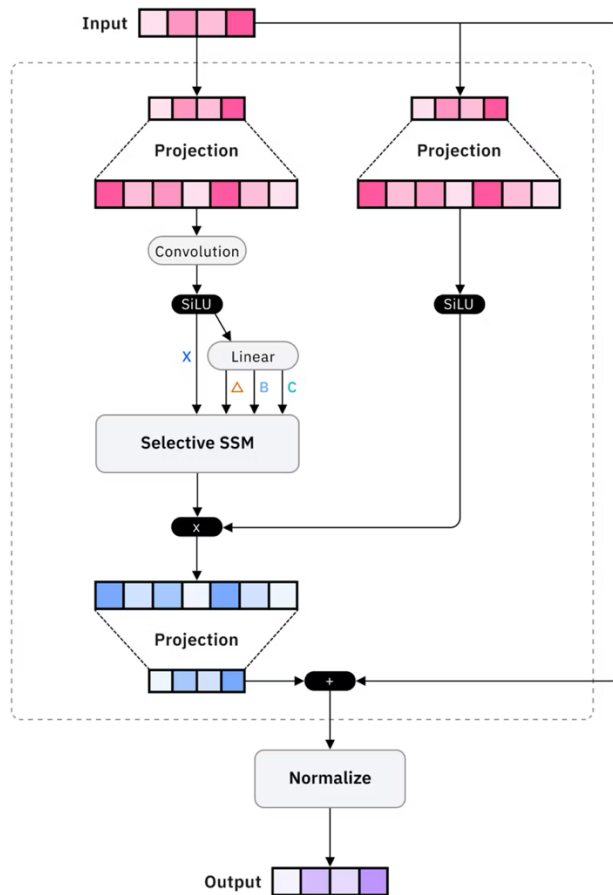
2: $\mathbf{B} : (\mathbf{B}, \mathbf{L}, \mathbf{N}) \leftarrow s_B(x)$ 3: $C : (B, L, N) \leftarrow s_C(x)$ 4: $\Delta : (B, L, D) \leftarrow \tau_{\Delta}(\text{Parameter} + s_{\Delta}(x))$ 5: $\bar{A}, \bar{B} : (B, L, D, N) \leftarrow \text{discretize}(\Delta, A, B)$ 6: $y \leftarrow \text{SSM}(\overline{A}, \overline{B}, C)(x)$

- **Time-varying**: recurrence (*scan*) only

7: **return** y

Mamba – Selective State Space Models (S6)

2. Mamba Block



Algorithm 2 SSM + Selection (S6)

Input: $x : (B, L, D)$

Output: $y : (B, L, D)$

1: $A : (D, N) \leftarrow \text{Parameter}$

▸ Represents structured $N \times N$ matrix

2: $B : (B, L, N) \leftarrow s_B(x)$

3: $C : (B, L, N) \leftarrow s_C(x)$

4: $\Delta : (B, L, D) \leftarrow \tau_\Delta(\text{Parameter} + s_\Delta(x))$

5: $\bar{A}, \bar{B} : (B, L, D, N) \leftarrow \text{discretize}(\Delta, A, B)$

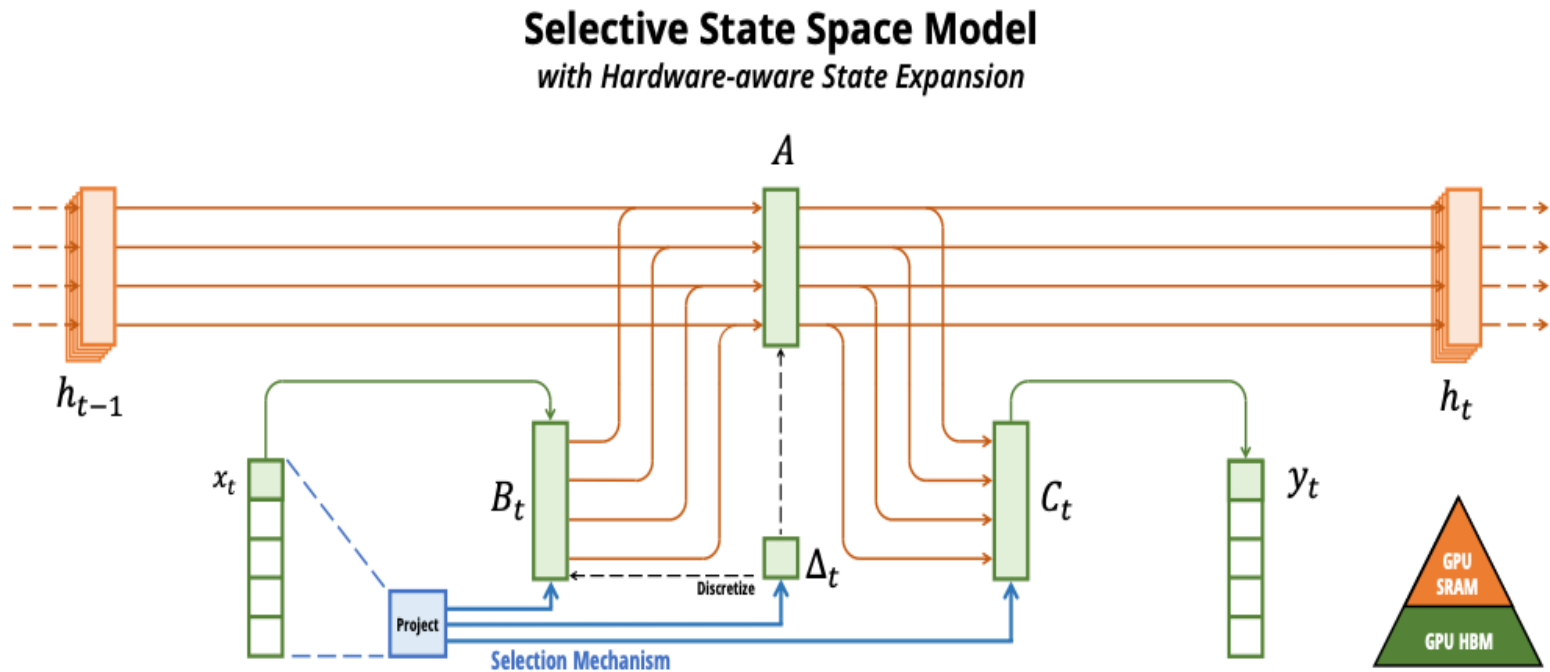
6: $y \leftarrow \text{SSM}(\bar{A}, \bar{B}, C)(x)$

▸ **Time-varying:** recurrence (*scan*) only

7: **return** y

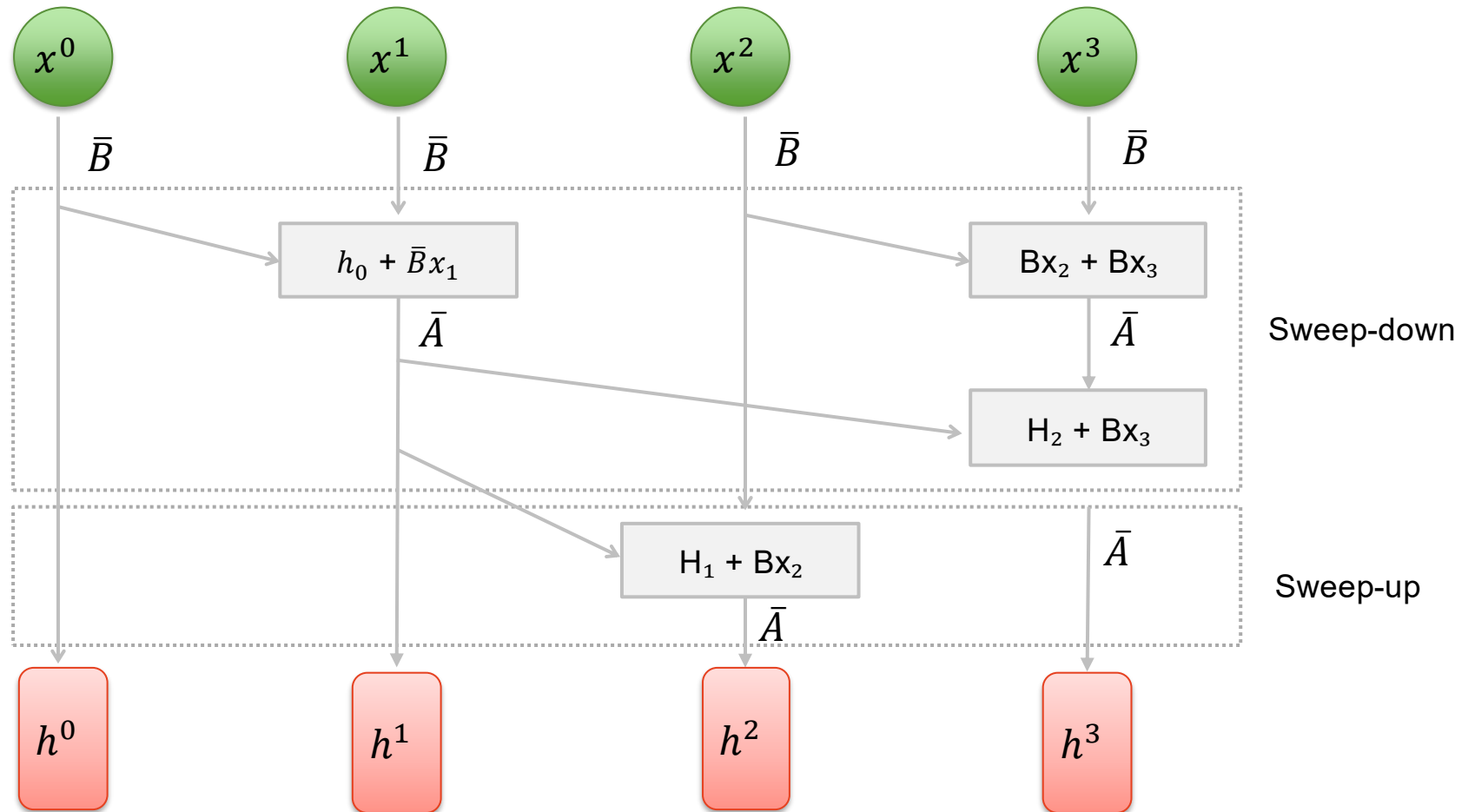
Mamba – Selective State Space Models (S6)

3. Hardware-aware State Expansion

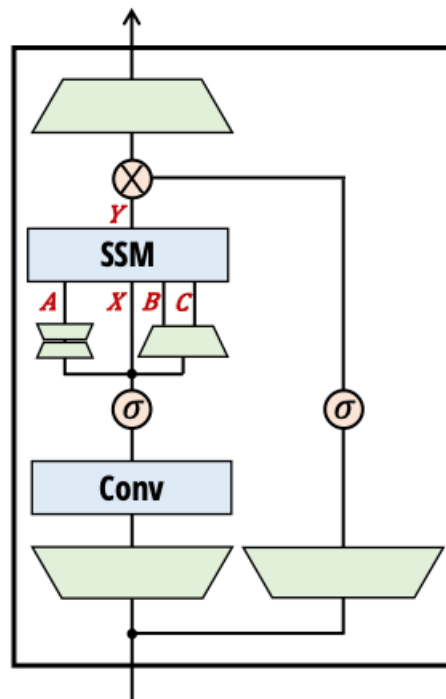


Mamba – Selective State Space Models (S6)

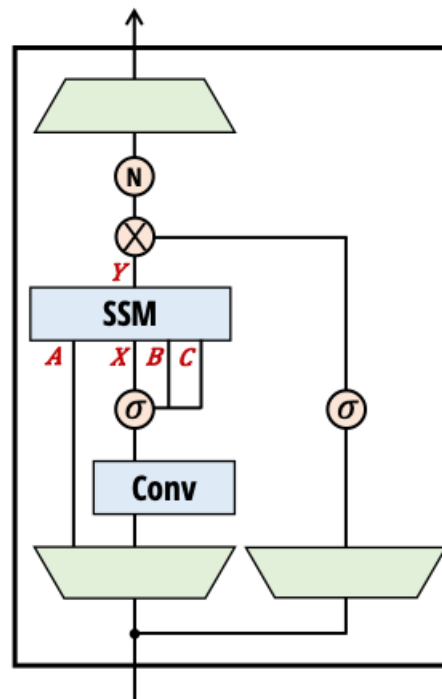
4. Parallel Scan



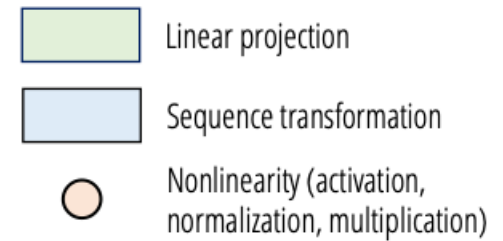
Mamba 2



Sequential Mamba Block

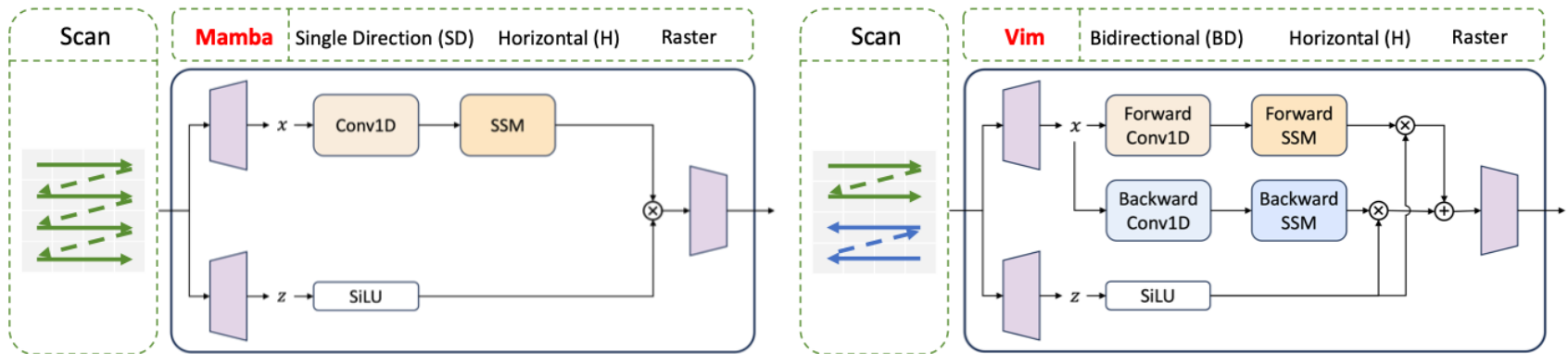
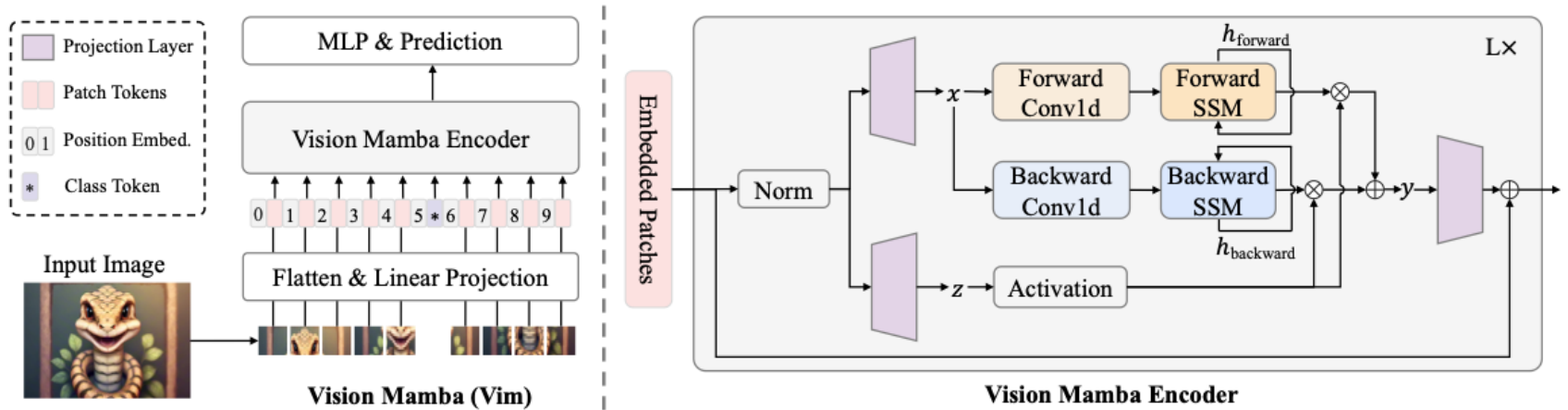


Parallel Mamba Block



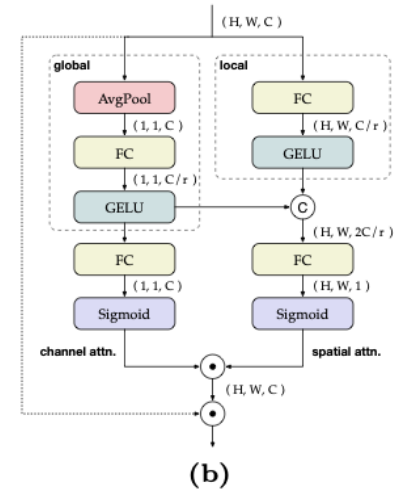
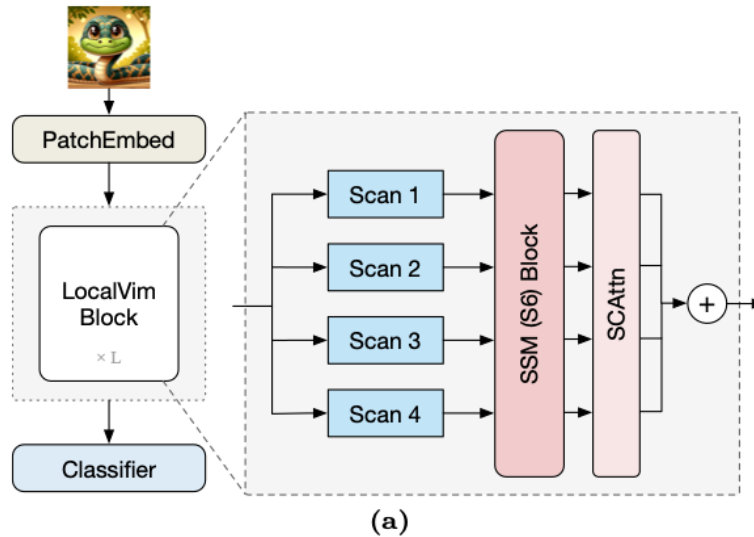
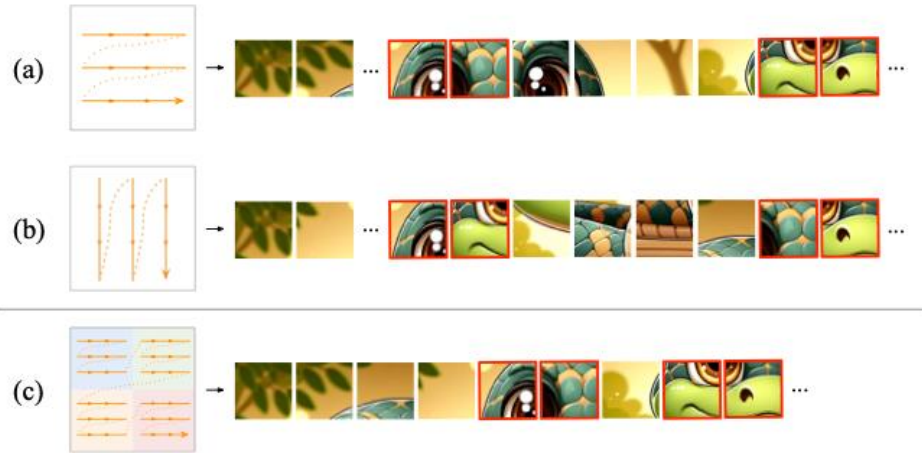
Mamba for Vision

Vision Mamba



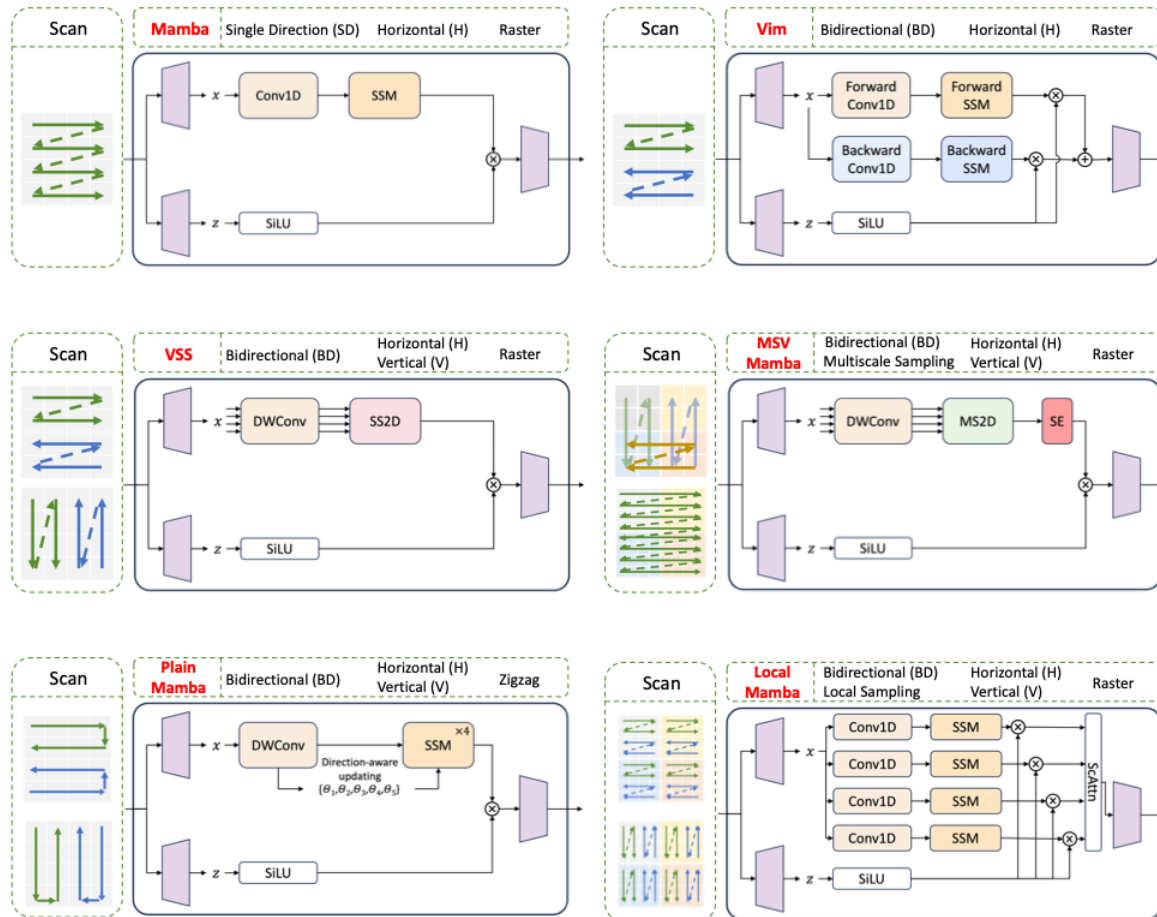
Mamba for Vision

LocalMamba



Mamba for Vision

Different Vision Mamba Blocks

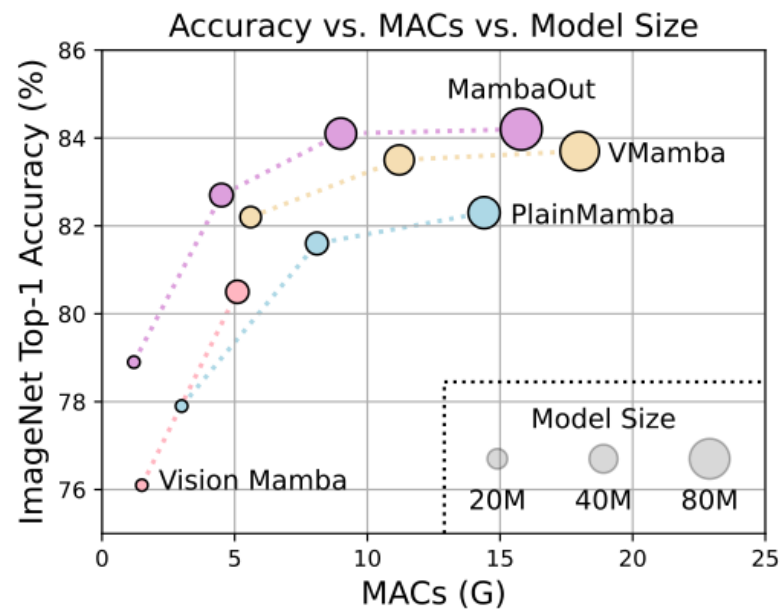
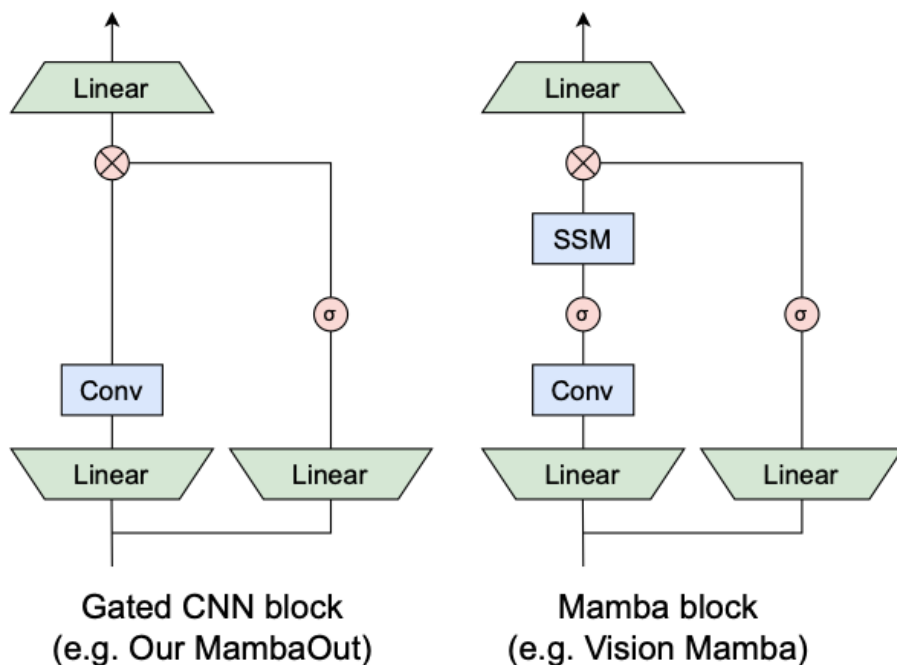


MambaOut: Do We Really Need Mamba for Vision?



In memory of Kobe Bryant

“What can I say, Mamba out.” — Kobe Bryant’s NBA farewell speech in 2016.



MambaOut: Do We Really Need Mamba for Vision?, CVPR 2025

Thank you !